Collective Trading Structure and Market Risk: From Aggregate Observations to Dynamic Inference

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Collective dynamics, inference and market risk

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Abstract

Can aggregate trading observations reveal collective structure that is useful for market risk? We develop a continuous mechanism-and-inference framework linking price–volume observations, effective trading weights, interacting decisions and future risk. Exact moments in a finite Erdős–Rényi reference identify parameters under known execution scales; ten independent recovery studies and exact observational-equivalence examples establish the scope of that inverse. Weighted covariance and temporal identities distinguish large collective decisions from raw execution counts and fragmented orders. A retained 48-unit fast–slow network supplies exact one-step flow moments, verified stochastic multistep propagation and paired experiments in which shared strategy signals and price feedback alter risk. An observable-only conditional simulation ensemble propagates uncertainty through common future flow/return paths and compares risk with equally supplied macro-statistical controls. The primary eight-step controlled QLIKE values are .13412 for the network ensemble and .13329 for the same-feature reference; paired uncertainty does not establish incremental accuracy. Bitcoin execution and time-scale diagnostics identify an uncalibrated observation layer before independent market testing. Historical volatility benchmarks, portfolio decompositions and strategy-overlap tools complete the application record. The results provide a reproducible route for testing which effective collective information is identifiable and whether it contributes to risk beyond directly observed pressure.

1 Introduction

Aggregate price and trading records compress many interacting decisions into returns, executed volume, counts and signed aggressive flow. A large decision can be executed in pieces; many independently acting traders can share a signal; and the same pressure can move prices differently as accommodation changes. The central question is whether a small observable interface can reveal risk-relevant collective structure, rather than merely reproduce a familiar market pattern.

The Cont–Bouchaud model links coordinated group actions to aggregate fluctuations[1]. Adaptive market models add switching and feedback[2], while calibration research distinguishes observable resemblance from parameter identification[6, 7]. We use these distinctions to connect a tractable finite-network reference to a weighted fast–slow decision model and an explicit observable-state inference task. A transaction price records an exchange under current information and trading constraints; our short-horizon model studies pressure and accommodation, not a universal intrinsic-value or eventual price-reversion law.

The research follows the sequence *macro observations → effective statistical structure → future joint trading/price paths → risk evaluation and application*. Exact moments and finite-ER inversion establish the initial inferential route. Weighted covariance and execution checks define what can be carried into market observations. The same interdependent network then supplies state dynamics, local perturbation response and common-signal/feedback experiments. Observable-only inference and common-path risk test the reverse route without supplying the target's true state or future.

Four questions organize the evidence: which structural quantities the reference identifies; which execution and temporal observations distinguish them; how the retained dynamic law turns state uncertainty into future risk; and whether that information improves an equally supplied forecast or reduces total measurement cost. Controlled simulations, Bitcoin observations and sector-portfolio

applications have distinct roles. The completed experiments provide conditional mechanism results, numerical propagation checks and prediction comparisons; market observation calibration and independent predictive confirmation are evaluated as separate requirements.

2 Finite network model and observable moments

2.1 Agents, actions, and the observation interface

Each period draws $H \sim G(N, p)$, where $p = \rho/(N-1)$, $N \geq 2$, and $0 \leq \rho \leq N-1$. Nodes are model traders, edges represent coordination, and a connected cluster C contains s_C traders. Each cluster shares $\xi_C \in \{-1, 0, 1\}$ with probabilities $q/2, 1-q, q/2$. Actions are independent across clusters and independent of the graph. Throughout, q is the total activity probability. The implementation uses the equivalent per-direction probability $a = q/2$.

Each active trader submits a unit order $u > 0$. A simplified external counterparty absorbs all submitted orders, and the log-price response is linear. The observations are

$$r = \frac{\lambda}{N} \sum_C s_C \xi_C, \qquad V = u \sum_C s_C |\xi_C|, \tag{1}$$
$$S_j = \sum_C s_C^j, \qquad \lambda > 0, \quad 0 < q < 1.$$

Signed price effects can offset, whereas trading activity adds. Here V counts executed activity and λ is a model response scale. Table 1 defines the notation, and Fig. 1 illustrates the observation rule.

Table 1. Model notation. A connected group shares one action within a period.

Symbol	Definition
N, ρ	Number of traders and expected ER degree, with $p = \rho/(N-1)$
C, s_C, S_j	Connected group, its size, and the power sum $S_j = \sum_C s_C^j$
q, ξ_C	Total activity probability and shared sell/inactive/buy action
u, λ	Unit order size and positive log-price response scale
r, V	Log return and total executed model volume per period
$A_N(\rho)$	Expected squared-size sum $\mathbb{E}S_2$ under the finite ER graph law

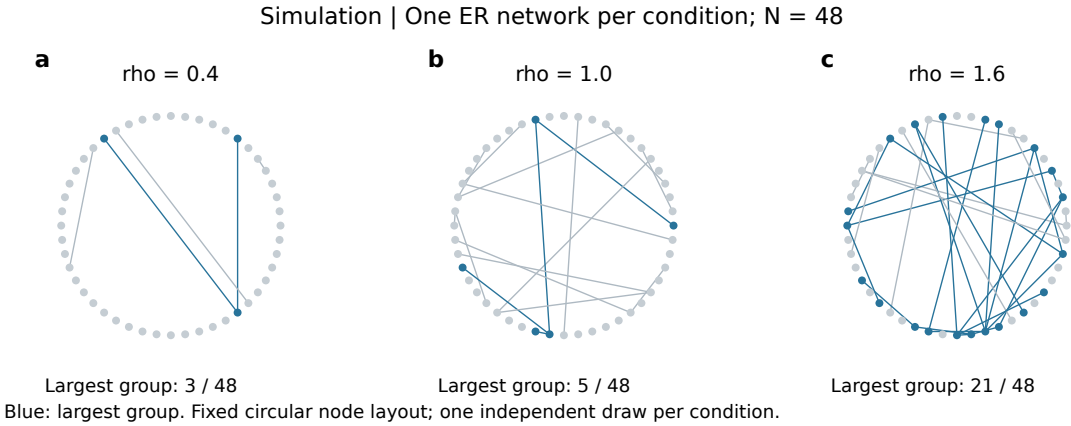


Figure 1. Shared group actions generate net price effects and total trading activity through the same execution rule. Nodes are model traders and connected groups coordinate their actions. [F01]

2.2 Exact moments and their mechanistic use

Zero-mean cluster actions eliminate cross terms in return variance. Conditional mean volume does not depend on the graph. Writing $A = \mathbb{E}S_2$ gives

$$\begin{aligned} \mathbb{E}V &= quN, \\ \mathrm{Var}(r) &= q(\lambda/N)^2 A, \\ \mathrm{Var}(V) &= q(1-q)u^2 A. \end{aligned} \tag{2}$$

Equal mean volume therefore need not imply equal risk. At fixed N, q, u, λ, a larger sum of squared cluster sizes increases return variance. For 100 independent traders, $S_2 = 100$; ten clusters of ten traders give $S_2 = 1000$. Variance differs tenfold in this model example while mean volume remains unchanged.

Higher-order relations retain the connection between collective activity and large price changes:

$$\begin{aligned} \mathrm{Cov}(r^2, V) &= u(\lambda/N)^2 q(1-q)\mathbb{E}S_3, \\ \kappa_3(V) &= u^3 q(1-q)(1-2q)\mathbb{E}S_3. \end{aligned} \tag{3}$$

Signed r and V have zero population covariance, while $|r|$ and V are positively associated under this interface. Price direction and magnitude thus require separate analysis. The empirical analysis below tests stronger joint restrictions than this association alone.

Conditional on the graph, the joint characteristic function is

$$\mathbb{E}[e^{itr+izV} \mid H] = \prod_C [1 - q + qe^{izus_C} \cos(t\lambda s_C/N)]. \tag{4}$$

It depends only on the multiset of cluster sizes, directly supporting the identification boundary below. For fixed N, $|r| \le \lambda$, so the mathematical tail is bounded. This bounded support sets the domain for the finite-range tail tests below.

Source tables and methods: `docs/theory/FOUNDATIONS.md`.

3 Percolation and finite return distributions

In the infinite sparse ER limit, degrees are Poisson(ρ) and $G_0(z) = G_1(z) = e^{\rho(z-1)}$. The edge-exploration generating function satisfies $H_1(z) = zG_1(H_1(z))$. A giant component emerges when the mean branching number crosses one. Its fraction satisfies

$$P_\infty = 1 - e^{-\rho P_\infty}, \qquad \rho_c = 1. \tag{5}$$

Below threshold, the expected cluster size of a uniformly selected node is $1/(1-\rho)$. In finite systems, S_2/N differs from the statistic excluding one largest component,

$$\chi_f = \frac{\sum_{C \neq C_{\max}} s_C^2}{\sum_{C \neq C_{\max}} s_C} \tag{6}$$

which is recorded as missing when no finite component remains.

We scanned $N = 128, 256, 512$, nine ρ values, and 40 graphs per cell, totaling 1,080 graphs. Figure 2 shows a smooth finite-size connectivity transition. Grid resolution and fluctuations affect peak positions. The infinite-system threshold organizes the finite-size comparison; the measured peak also depends on system size and grid resolution.

The mean-field critical law $n_s \propto s^{-5/2}$ concerns cluster number density. Selecting a node uniformly instead weights cluster sizes by s. Returns sum signed cluster actions; activity, cancellation, and bounded support affect their tails. For density $f(x) \propto x^{-\alpha}$, the complementary cumulative distribution (CCDF) exponent is $\alpha - 1$. These different sampling and aggregation operations require separate exponent definitions.

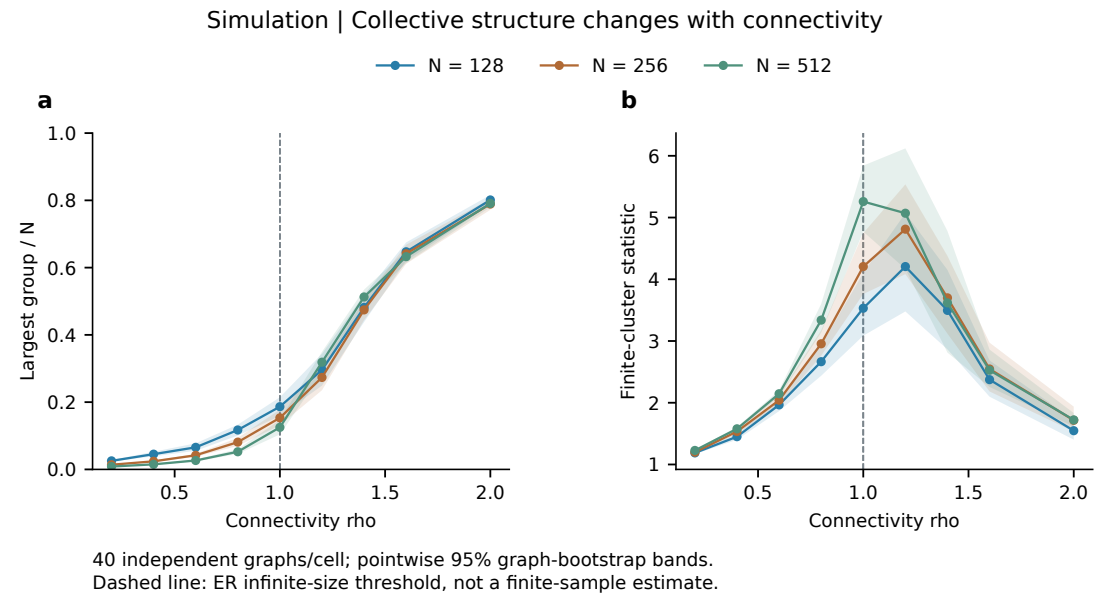


Figure 2. Finite-size connectivity transition for three population sizes. Each N/ρ cell contains 40 independent graphs. The vertical line marks the infinite sparse-ER threshold; finite-system peaks depend on size and grid resolution. Source: simulation_study_v1/phase_summary.csv. [F03]

Five conditions each generated 8,192 periods, totaling 40,960 simulated intervals. Over the prespecified absolute net-order range $m = 4, \ldots, 128$, all three ρ conditions rejected the simple finite-support pure power law. The raw $p = 0.005$ is the resolution floor of 199 parametric-bootstrap draws; Holm adjustment across three tests gives $p = 0.015$. A better relative AIC under some conditions does not imply adequate absolute fit. The implementation follows maximum-likelihood, Kolmogorov–Smirnov (KS), and alternative-distribution principles[4], adapted to finite integer support.

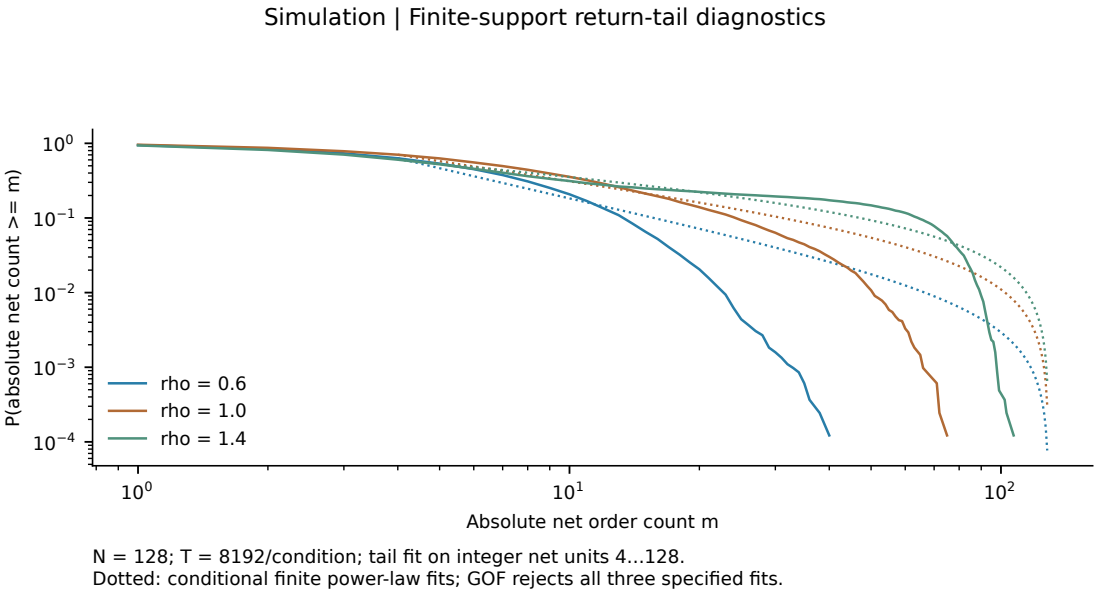


Figure 3. Return CCDFs and finite-support tail fits over the stated net-order range. The pure-power-law goodness-of-fit test rejects all three fitted connectivity conditions; relative AIC comparisons measure a separate property. [F05]

Collective amplification therefore survives even where this finite return distribution fails a pure-power-law fit. Cluster geometry and return tails answer different questions. Appendix B compares cluster distributions, population-size changes, and temporal aggregation.

4 Parameter recovery and observational equivalence

4.1 Finite-graph moment inversion

For known N, u under the fixed observation assumptions, population moments give

$$q = \frac{\mathbb{E}V}{uN}, \qquad A_N(\rho) = \frac{\mathrm{Var}(V)}{q(1-q)u^2}, \tag{7}$$
$$\lambda^2 = N^2u^2(1-q)\frac{\mathrm{Var}(r)}{\mathrm{Var}(V)}.$$

The ER mapping $A_N(\rho) = \mathbb{E}S_2$ is strictly increasing and therefore supports inversion. Let $C_s(p)$ be the probability that $G(s,p)$ is connected, and let $\pi_{N,s}$ be the probability that a uniformly selected trader belongs to a group of size s.

At fixed p, Gilbert's connectivity recursion is[5]:

$$C_s = 1 - \sum_{k=1}^{s-1} \binom{s-1}{k-1} C_k (1-p)^{k(s-k)},$$
$$\pi_{N,s} = \binom{N-1}{s-1} C_s (1-p)^{s(N-s)}, \tag{8}$$
$$A_N(\rho) = N \sum_s s\pi_{N,s}, \qquad C_1 = 1.$$

This classical recursion supplies the finite-size map needed by the moment estimator. The identity $A_N = N + N(N-1)\Pr(1 \leftrightarrow 2)$ connects it to pairwise connectivity. Coupling graphs by adding edges makes this probability strictly increasing in p for $N \geq 2$, which gives the population inverse its uniqueness. The implementation used 380 decimal digits, checked against higher precision and small-graph enumeration.

Ten independently regenerated studies produced 4,920 windows of 256 periods, totaling 1,259,520 simulated intervals. Each study used 252 training, 84 validation, and 84 test windows, plus 72 unseen intermediate-parameter windows. Moment inversion and classical radial-basis-function (RBF) methods share evaluation units. The return-only view contains return standard deviation, mean absolute return, the 95th and 99th absolute-return quantiles, and the zero-return fraction. The volume-only view contains volume mean, standard deviation and 95th percentile. Joint inputs combine these views and the Pearson correlation of volume with absolute return. Seeds and label metadata are excluded.

The supervised reference is kernel ridge regression with a Gaussian radial-basis kernel. For standardized observed features x, training labels $y_i = \rho_i$, and training mean $\bar{y}$,

$$K_{ij} = \exp[-\gamma\|x_i - x_j\|^2],$$
$$\widehat{\rho}(x) = \bar{y} + k(x)^\top (K + \alpha_K I)^{-1}(y - \bar{y}\mathbf{1}). \tag{9}$$

Here $k_i(x) = \exp[-\gamma\|x - x_i\|^2]$ and α_K is the ridge penalty. Median imputation, missingness indicators and standardization are fitted on training windows only. Validation MAE selects the penalty and kernel width from the fixed configuration grid. For positive return standard deviation, the scale-invariant view divides the three return-magnitude summaries by that scale and retains the zero fraction and volume summaries; degenerate windows follow the imputation rule. Table 2 and Fig. 4 report the matched-scale results.

On the original grid, joint RBF reduced average MAE by 44.49% relative to return-only RBF. Volume-only RBF performed better than joint RBF in all ten studies. For unseen parameters, moment inversion achieved MAE 0.027114, compared with 0.026882 for joint RBF. Mechanistic formulas can therefore provide interpretable estimators within the specified family.

4.2 Price rescaling separates representation error from information loss

Each window undergoes $r \mapsto br$, with $b = 0.5, 1, 2$, while V remains unchanged. These are paired transformations of the same windows. Doubling the price response increases raw joint-RBF MAE

Table 2. Connectivity recovery in the specified ER family: mean MAE over ten independent studies, in units of ρ.

Method	Original grid	Unseen intermediate ρ
Return-only RBF	0.048660	0.045077
Joint RBF	0.027013	0.026882
Volume-only RBF	0.022196	0.036849
Scale-invariant joint RBF	0.032106	0.031013
Moment inversion	0.028372	0.027114

Each study regenerates training, validation, and evaluation windows. Source: depth_study_v1/summary_metrics.csv.

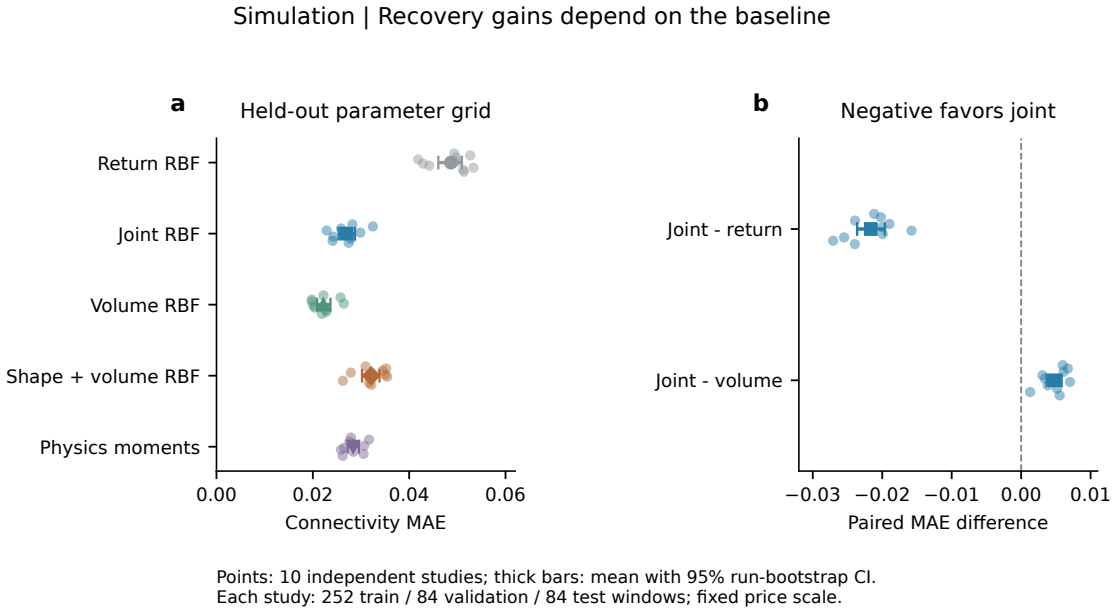


Figure 4. Parameter recovery across ten independently regenerated and retrained studies. Points are studies; summaries and intervals describe across-study variation. Table 2 gives grid and unseen-parameter MAE under the matched-scale design. [D01]

from 0.027013 to 0.258133. The scale-invariant representation and moment method remain at 0.032106 and 0.028372, respectively. Figure 5 shows this representation trade-off.

4.3 Even complete observations need not reveal internal edges

The chain $\{12, 23, 34\}$ and star $\{12, 13, 14\}$ differ as four-node graphs, yet both form a single cluster of size four. Equation (4) gives identical complete joint price–volume laws. More passive observations of this kind cannot recover the internal edges. This is stronger than failure of a particular classifier. Exact certificates for general cluster mixtures are also included with proofs and rational checks.

Parameter recovery, structural equivalence, and information lost through feature compression require separate treatment. Financial ABM calibration studies likewise distinguish reproducing stylized facts from identifying mechanism parameters[6, 7]. The characteristic function, known-truth recovery experiments, and market restrictions thus address structural information, estimation accuracy, and model compatibility, respectively.

Source tables and methods: `docs/theory/DEPTH_ADDENDUM.md`.

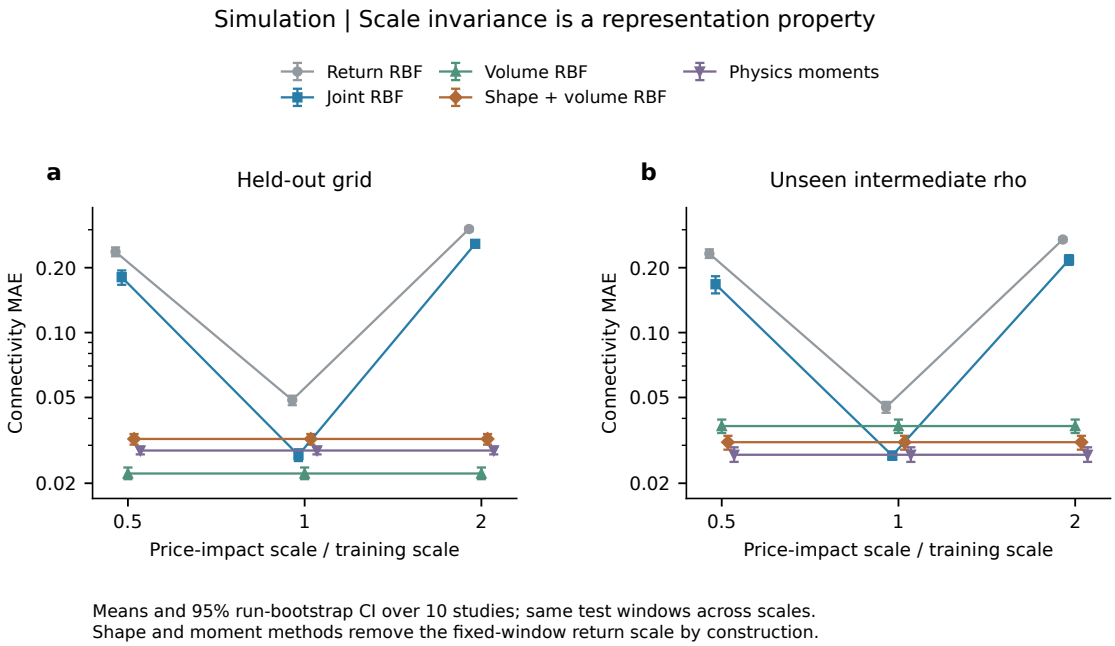


Figure 5. Price-response rescaling applied to the same evaluation windows in ten studies. Raw-return inputs change with scale, while moment inversion and scale-invariant inputs remove the specified window-level transformation by construction. Intervals resample studies. [D02]

5 Testing joint observations in Bitcoin

5.1 Data and price–volume comparison

The primary market case uses Binance Vision BTCUSDT spot five-minute bars[10]. The 2025 sample contains 105,120 bars and 365 complete UTC days, without filling or deletion. H1 provides calibration; H2 contains the original 184-day compatibility-test period. Returns are adjacent-close log changes, and bar volume is measured in BTC. Five-minute bars still compress the within-bar trade sequence. Figure 6 compares binned joint statistics and price-response proxies using total BTC trading volume.

Appendix Fig. 21 compares market and model tails on a calibration-defined scale. The market bars record price and volume, while trader counts, clusters and ρ remain unobserved. The empirical analysis therefore tests observable moment restrictions directly.

5.2 A joint necessary restriction derived from the mechanism

Eliminating scales from Eq. (3) gives a restriction for any graph distribution, beyond ER:

$$\Delta = \frac{1}{\mathbb{E}V}\left[\frac{\mathrm{Cov}(r^2, V)}{\mathrm{Var}(r)} - \frac{\kappa_3(V)}{\mathrm{Var}(V)}\right] = \frac{\mathbb{E}S_3}{N\mathbb{E}S_2} \in [1/N, 1]. \tag{10}$$

It requires fixed N, q, u, λ, symmetric independent actions, simplified execution, and linear response. Empirical calculations remove constant return drift and retain higher-moment and nonstationarity sensitivities.

For market H2, $\Delta = -8.181$, with an approximate 95% circular seven-day-block interval $[-11.998, -4.712]$. Figure 7 contrasts this estimate with the required positive range. The observed sign conflicts with the necessary range under the joint fixed-interface assumptions. All 12 calendar-month point estimates in 2025 are also negative, as are the checks excluding the highest-volume or highest-realized-variance day. These source-table sensitivities support using the restriction as a model diagnostic. The test identifies a failure of the fixed interface, while the next section examines how changing activity and response affects that restriction.

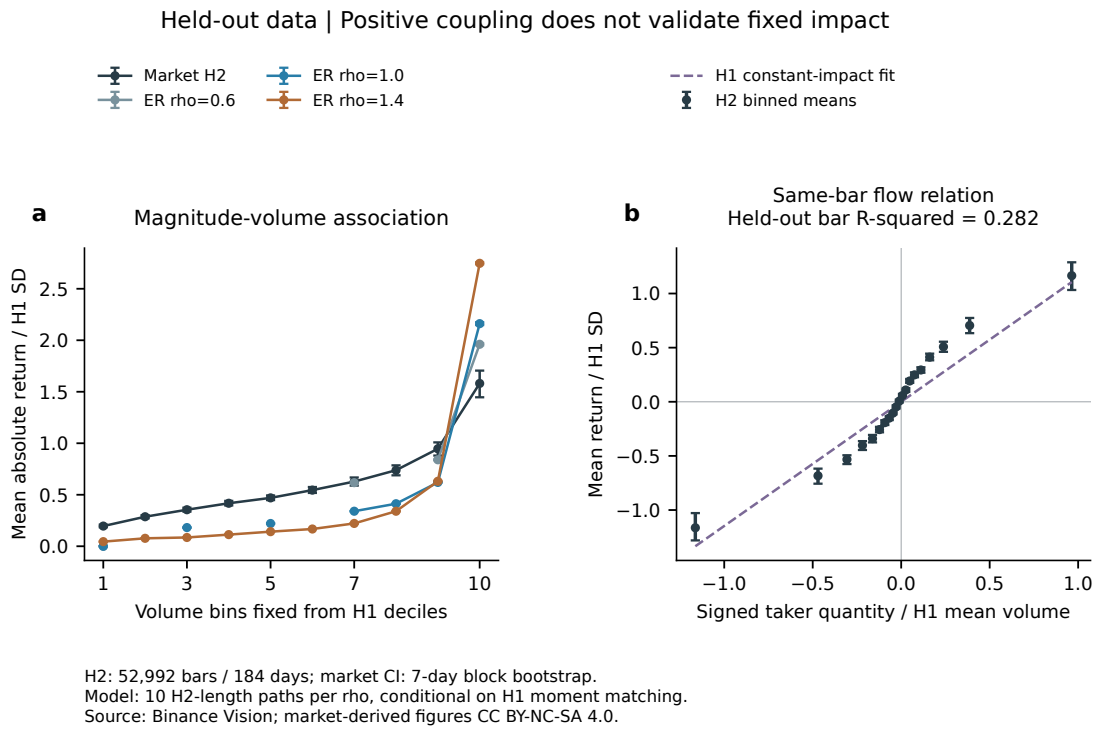


Figure 6. Bitcoin and model price–volume comparisons using H1-fixed bins. H2 magnitude–volume Spearman correlation is approximately 0.498, or 0.481 after seasonal adjustment. Empty model bins remain missing. Binance Vision, CC BY-NC-SA 4.0. [R01]

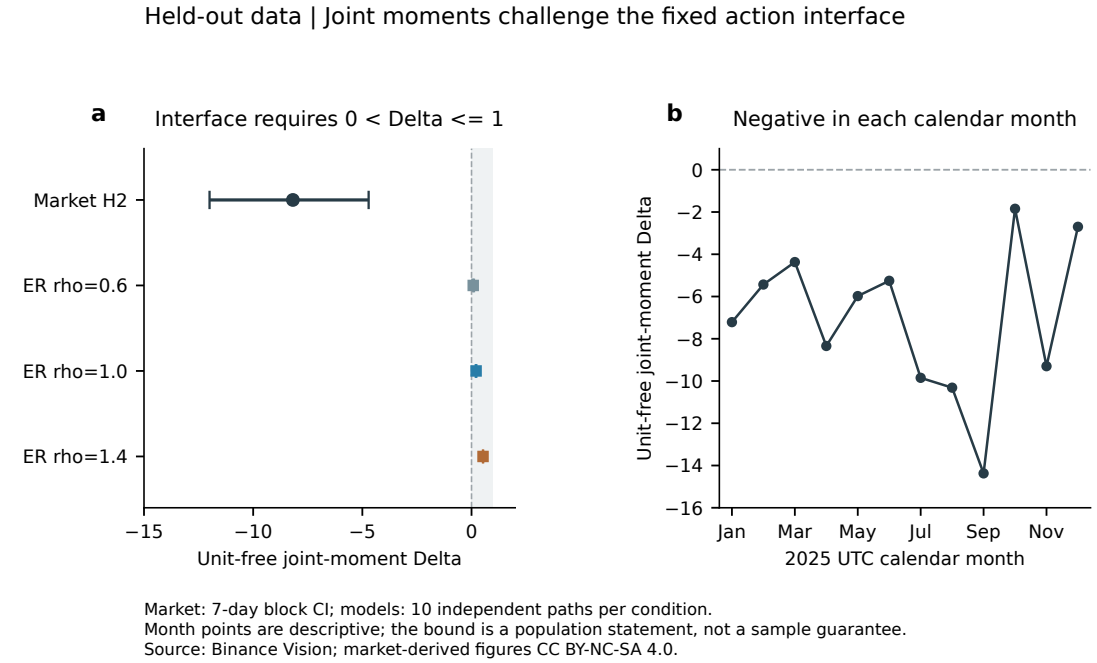


Figure 7. Joint-moment restriction for the fixed group-action interface and its BTC H2 estimate. The model requires a positive statistic. Market intervals use circular date-block resampling; monthly and extreme-day sensitivities are retained in the source tables. Binance Vision, CC BY-NC-SA 4.0. [R04]

5.3 Temporal structure supplies additional forecasting information

Let $\mathcal{F}_t$ denote the information available through period t. With fixed parameters and independently redrawn graphs and actions,

$$\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = q(\lambda/N)^2 \mathbb{E}S_2 \tag{11}$$

is constant. This static baseline does not itself provide a date-varying next-period risk forecast. Market H2 has lag-one magnitude autocorrelation 0.344, signed-return autocorrelation about

−0.021, and a one-day aggregation variance ratio about 0.749. Temporal information therefore supplies an additional axis of mechanism comparison beyond the same-period cluster geometry.

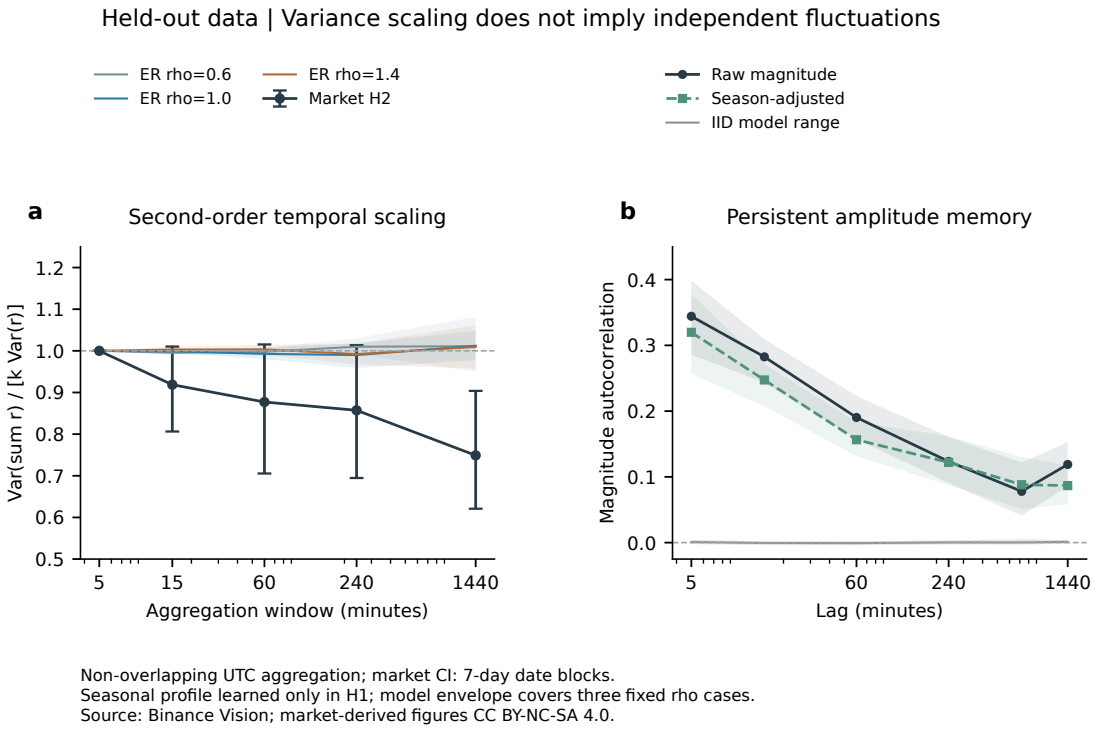


Figure 8. Temporal aggregation and lag dependence in independent model simulations and Bitcoin observations. Market magnitude memory contrasts with the independent-period baseline. Variance ratios and lag correlations describe temporal information, separately from cluster geometry. [R03]

6 Temporal states and forecasting information

To isolate temporal information, the controlled state extension keeps $N = 128, \rho = 1$ and independent graphs, while activity q_t or response λ_t follows an exogenous temporal state. Four arms each use ten paths of 8,192 periods, totaling 327,680 observations. Population mean volume and return second moments match across arms. Persistent activity produces magnitude and volume memory; persistent independent response can produce magnitude memory with nearly absent volume memory. Shuffling complete (r, V) pairs preserves the same-period empirical joint distribution while removing temporal memory. Figure 9 separates these informational roles.

Let $m_k = \mathbb{E}(q^k)$ and $A = \mathbb{E}S_2$, with activity independent of the graph. The total-variance identity gives

$$\mathrm{Var}(V) = u^2[(m_1 - m_2)A + N^2(m_2 - m_1^2)]. \tag{12}$$

The second term arises from changing activity, preventing interpretation of all volume variance as cluster structure. A complementary analytical construction, developed after the simulation comparison, sets $\lambda(q)^2 = \lambda_0^2 \bar{q}/q$, where $\bar{q} = m_1$ is mean activity. This keeps $\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = (\lambda_0/N)^2 \bar{q} A$ constant. Volume may reveal activity states without improving this forecasting target. Its population $\Delta \approx -2.16348$ shows that the positive fixed-interface restriction does not extend to all herding mechanisms. This construction demonstrates how activity–response coupling can reverse the restriction; estimating such a mechanism in market data remains a separate inference task. Full assumptions and exact checks appear in STATE_MECHANISM_v2.md.

An observable can identify the current activity state while its usefulness for forecasting return variance depends on state persistence and response compensation. This distinction motivates evaluating temporal observables in the forecasting benchmark.

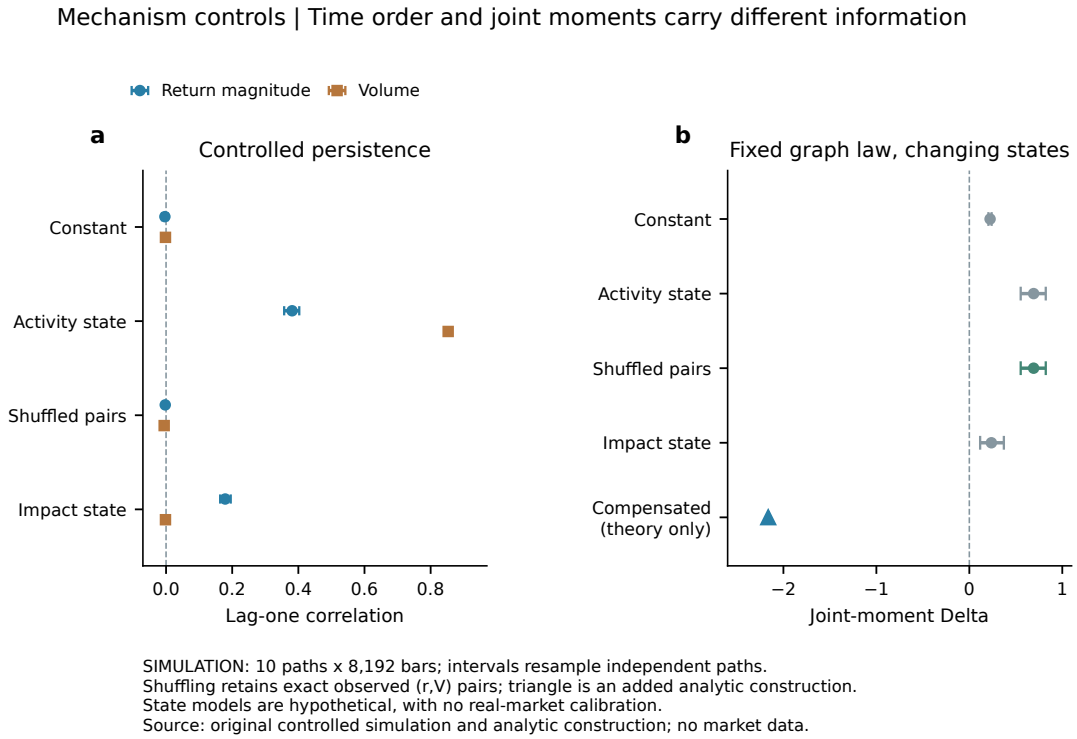


Figure 9. Controlled activity and response states under the same graph law. Persistent states produce different magnitude and volume memory; pair shuffling preserves the empirical same-period joint distribution and removes temporal order. Intervals resample independent paths. Sources: state_mechanism_v2. [V01]

7 Effective trading weights and the observable risk interface

The static group model supplies a tractable reference for a more general question: which statistical features of collective demand are needed for risk? Replace group size by a positive effective trading weight Q_C. A large decision, several investors acting together, and one parent order executed in pieces can contribute similar aggregate pressure. Their participant interpretations differ, so effective weights are defined in units of trading quantity rather than people.

7.1 The covariance mechanism survives the extension

For action vector ξ and fixed weight vector Q,

$$F = Q^{\mathsf{T}}\xi, \tag{13}$$

$$\begin{aligned}\mathrm{Var}(F \mid Q) &= Q^{\mathsf{T}}\Gamma Q \\ &= \sum_C Q_C^2 \,\mathrm{Var}(\xi_C \mid Q) + 2\sum_{C<D} Q_C Q_D \,\mathrm{Cov}(\xi_C, \xi_D \mid Q),\end{aligned} \tag{14}$$

where $\Gamma = \mathrm{Cov}(\xi \mid Q)$. The original independent symmetric-group law is the special case $\mathrm{Var}(F \mid Q) = q\sum_C Q_C^2$. With positive weights and marginal action variances fixed, additional positive cross-action covariance increases flow variance. Coupling can also change activity, mean direction, temporal dependence and price accommodation; its full price-risk effect is evaluated separately.

For execution sizes v_i in one bar, define the independent-sign reference

$$B = \sum_i v_i^2, \qquad G_V = \frac{\mathrm{Var}(F)}{B}, \qquad G_E = \frac{\mathbb{E}F^2}{B}. \tag{15}$$

The averaging window and execution unit must be declared. G_V is centered dispersion; G_E also contains mean-flow energy. Neither is a measured fraction of followers. Under the fixed symmetric sign reference, B is a size baseline; changing the grouping of executions changes that baseline.

A common-sign 100-unit parent split into ten 10-unit children has cumulative-flow variance 10,000 under a symmetric parent sign. Ten independent signs give 1,000. Both constructions have sum of squared child flows 1,000. In general,

$$\operatorname{Var}\left(\sum_t F_t\right) = \sum_t \operatorname{Var}(F_t) + 2\sum_{s<t} \operatorname{Cov}(F_s, F_t). \tag{16}$$

Thus directional persistence matters for cumulative risk, while it need not raise expected realized variance (RV) when per-bar second moments and a linear price response are fixed. This distinction determines the forecast target rather than serving as a cosmetic choice of risk statistic.

The execution bridge also yields a conditional inverse with heterogeneous individual sizes. Under independent symmetric group actions with common activity q, let i.i.d. positive individual marks have moments μ_1, μ_2, independently of the group structure and actions. Each active individual supplies one execution. For $\chi = \mathbb{E}S_2/N$,

$$G_E = 1 + \frac{\mu_1^2}{\mu_2}(\chi - 1), \qquad \chi = 1 + \frac{\mu_2}{\mu_1^2}(G_E - 1). \tag{17}$$

With equal marks, the amplification ratio equals group susceptibility. Exact finite enumeration and 300 independent simulation replications comprising 1.5 million periods verify the marked-size correction under its specified execution map. The map itself is identifying information: one participant splitting a two-unit parent into two unit fills and two participants sharing one common action can have the identical full (F, V, K, r) event law. The inverse concerns an effective group statistic under the execution assumptions, not a universally recoverable head count.

7.2 What the public execution data actually constrain

The measurement audit covers 663 complete Bitcoin days (190,944 five-minute bars) and 24 dates with microscopic labels. Raw and exchange-aggregate records reconcile with bar totals. Three earlier fixed-date cases contain 12,536,460 raw executions and 4,378,760 aggregate units. Exchange aggregation is an observed execution grouping, not complete parent-order reconstruction.

Nonnegative sizes imply

$$\frac{V^2}{K} \le B_{\text{raw}} \le V^2. \tag{18}$$

The mean ratio of these arithmetic bounds is about 22,754 on the complete daily records. Macro V, K alone therefore leave a broad size-concentration range. On sampled H2 dates, aggregate-execution B is on average 2.391 times raw-execution B. A January-only Gamma size calibration gives inferred energy amplification 45.85 versus observed 41.71 in July, but 56.39 versus 25.80 in October. The latter comparison exposes transfer error in the size baseline.

For a declared execution unit e with independent Gamma sizes of shape a_s, conditional normalized sizes are Dirichlet. Given its execution count $K^{(e)}$,

$$\mathbb{E}(B^{(e)} \mid V, K^{(e)}, a_s) = V^2 \frac{a_s + 1}{K^{(e)} a_s + 1}. \tag{19}$$

The working state rule targets exchange-aggregate B, predicting aggregate counts from the observed raw count through an H1-calibrated count map and using state-conditioned concentration. The fitted Gamma shape is a prediction parameter, not evidence that empirical sizes are i.i.d. Gamma. Relative to a refitted pooled-Gamma control, StateGamma reduces aggregate-B loss from .2691 to .2549 (5.28%). The 24-date audit reconciles approximately 88.1 million raw executions and 29.2 million aggregate units from 48 checksum-verified archives.

Across 12 held H2 microscopic-label dates spanning six months, mean daily absolute relative amplification error is 35.14%; providing observed aggregate counts reduces it to 18.18%, at higher information cost. Measured and inferred G_E share a numerator, so these discrepancies evaluate the execution baseline, not independent flow-variance recovery. Centered and uncentered statistics are reported separately.

Large count dispersion also has alternatives to coordination. In 256 independent repetitions per null construction, heterogeneous-intensity Poisson processes without self-excitation produce high Fano ratios. Their representative values, about 2,802 and 7,091 for varying-intensity and mixed-state constructions, coexist with a real count ratio near 13,784. A near-critical value obtained by substituting count moments cannot be identified with a market branching ratio or follower fraction.

Evidence record: measurement audit W1 and the fixed-date execution and macro-estimator amendments.

7.3 The risk map and its predictive requirement

For the original equal-size interface, $Q_C = us_C$ gives $r = [\lambda/(Nu)]F$, so the linear response below recovers its price law. For information $\mathcal{I}_t$, write $m_F = \mathbb{E}(F \mid \mathcal{I}_t)$ and a local response $r = \alpha + \beta F + \eta$. If α, β are fixed conditional on $\mathcal{I}_t$, $\mathbb{E}(\eta \mid \mathcal{I}_t) = 0$, and $\mathrm{Cov}(F, \eta \mid \mathcal{I}_t) = 0$, then

$$h = \mathbb{E}(r^2 \mid \mathcal{I}_t) = (\alpha + \beta m_F)^2 + \beta^2 \overline{B} G_V + \sigma_\eta^2. \tag{20}$$

This connects the group covariance mechanism to a return second moment. A same-window product $\overline{B}G_V$ is exactly the flow variance used to define it. Forecast information must therefore come from future state evolution, a cheaper valid measurement route, or a tested restriction that improves an information-matched forecast. Re-labelling a measured variance does not establish that increment.

8 Interdependent decisions, perturbations and future distributions

8.1 A weighted fast–slow decision network

The dynamic extension retains 48 effective units with fixed positive normalized weights w_i, two fixed sparse graphs and binary fast/slow states s_i^f, s_i^s. The fast layer responds to current neighbor pressure; the slow layer supplies a more persistent decision state. Each unit also carries an adaptation trace z_i. Interconnected opinion processes motivate this layered construction[13]; its financial interpretation and numerical rules are stated and tested here.

With capital-weighted neighbor direction u_i^h, the fields are

$$\begin{aligned} H_i^f &= J_f u_i^f + \kappa s_i^s + \rho_f \mu s_i^f - a_f \mu z_i + h_t, \\ H_i^s &= J_s u_i^s + 0.55\kappa s_i^f + \rho_s \mu s_i^s - a_s \mu z_i + 0.25 h_{\mathrm{exec},t}. \end{aligned} \tag{21}$$

The next fast sign is sampled with probability $[1 + \tanh(1.5H_i^f)]/2$ of being positive. The slow state attempts an update with probability 0.14 per step, and

$$z_i' = (1 - \epsilon) z_i + \epsilon s_i^{f,\mathrm{new}}, \qquad \epsilon = 0.055. \tag{22}$$

Here J_f, J_s are within-layer influence strengths, κ is cross-layer coupling, ρ_f, ρ_s are inertia coefficients and a_f, a_s are adaptation coefficients; these layer coefficients are distinct from the ER degree ρ. The inertia term favors retaining a direction; the adaptation term can reduce the attractiveness of a persistent orientation. Opposing positions are signs within this model. Layer coupling is same-unit fast/slow influence, not a measured institution-to-institution dependency or a rewiring rule.

Independent activity draws A_i with probability $a = 0.58$ generate background quantities $C_0 w_i A_i$, with $C_0 = 1200$. For an additional deterministic pulse d,

$$F = d + C_0 \sum_i w_i A_i s_i^{f,\mathrm{new}}, \qquad V = |d| + C_0 \sum_i w_i A_i. \tag{23}$$

The model count $K = \sum_i A_i$ records active effective units. It is not the number of exchange executions. A positive accommodation state L and source noise convert flow into return through the frozen source law,

$$r = \mathrm{clip}\left(0.009 \tanh\left[\frac{0.34(F/1200)}{L} \right] + \eta, -0.018, 0.018 \right). \tag{24}$$

Accommodation is a simulated response state, not an observed order book. This law allows both flow alignment and absorption to affect price fluctuations. It does not contain an estimated intrinsic-value process.

8.2 A perturbation can induce trading without increasing every risk target

Paired-seed interventions compare the same background innovations with and without a pulse. Across 128 paired seed units, a 24-unit buy pulse raises first-step flow by 33.817 on average, with 95% paired interval $[30.062, 37.987]$. Subtracting the direct pulse leaves an induced background response of about 9.817 units. Thus the model gives a concrete route from a large trading decision to additional aligned trading.

The 48-step RV difference has mean -1.50×10^{-7} and interval $[-4.94\times10^{-7}, 1.36\times10^{-7}]$. At eight fixed latent validation states, the same positive pulse raises next-step flow second moment at four states and lowers it at four: a buy pulse can partly offset an initially negative state. These results distinguish a local response from a universal volatility increase. Without a scheduled opposite parent program, 108 of 128 seeds cross between opposite alignment thresholds. Such random internal switching is a model outcome, while strategic early exit and an inevitable consensus-to-crash cycle remain separate hypotheses.

8.3 Exact one-step moments and distribution-preserving propagation

Let $t_i = \tanh[1.5H_i(S, d)]$ for complete state S, with the common-input channel off. Independent transition and activity draws give

$$m_F(S, d) = d + C_0 a \sum_i w_i t_i, \tag{25}$$

$$v_F(S, d) = C_0^2 \sum_i w_i^2 (a - a^2 t_i^2). \tag{26}$$

When a common future input is present, these formulas are conditional on that input, which must subsequently be integrated. Exact conditional F, V, K means and covariances provide the observation interface. One-step Monte Carlo agrees with the formulas, and a 144-dimensional conditional-mean Jacobian agrees with finite differences to a maximum error of about 1.62×10^{-10}.

Repeated substitution of the mean state is a different operation: in a nonlinear model, $\mathbb{E}[f(S)]$ need not equal $f(\mathbb{E}S)$. A pair of hidden arrangements can have equal current F, V, K and aggregate fast/slow directions but different next-step mean flows, 412.47 and 566.37 in a retained example. Those current summaries are therefore insufficient for exact closure.

The operational solution propagates complete stochastic states and uses their joint future paths. A verification design uses eight new latent states and three pulse conditions, with 12,288 prediction paths, 98,304 independent reference paths and 196,608 source-engine one-step branches. For the unpulsed 48-step flow mean, RMSE normalized by background quantity is 0.00785 for stochastic propagation and 0.38842 for mean iteration (Fig. 10). These are numerical propagation errors against an independent simulation reference, not forecast losses on market data. The result supplies a tested multistep continuation of the original network law.

Evidence record: network perturbation W2; exact mathematical interface and propagation D01.

9 Shared strategy signals and feedback on the same network

Similar inputs across trading strategies give an economic source of correlated decisions. Rather than attaching a separate predictor, we add a common signal to the retained decision fields and compare it with independent node signals having identical marginal distributions. Positive and negative lagged price feedback, together with a common activity rule, are tested as distinct channels.

The 2007 quant episode provides an economic motivation: factor-portfolio and transaction evidence is consistent with overlapping deleveraging and temporary withdrawal of market-making capital[16]. Funding/market-liquidity theory also supplies a conditional feedback channel between the ability to finance positions and the ability to absorb trades[17]. These precedents motivate separate shared-input and response mechanisms. Our experiment does not infer actual institution identities, reconstruct their portfolios or implement their complete margin rules.

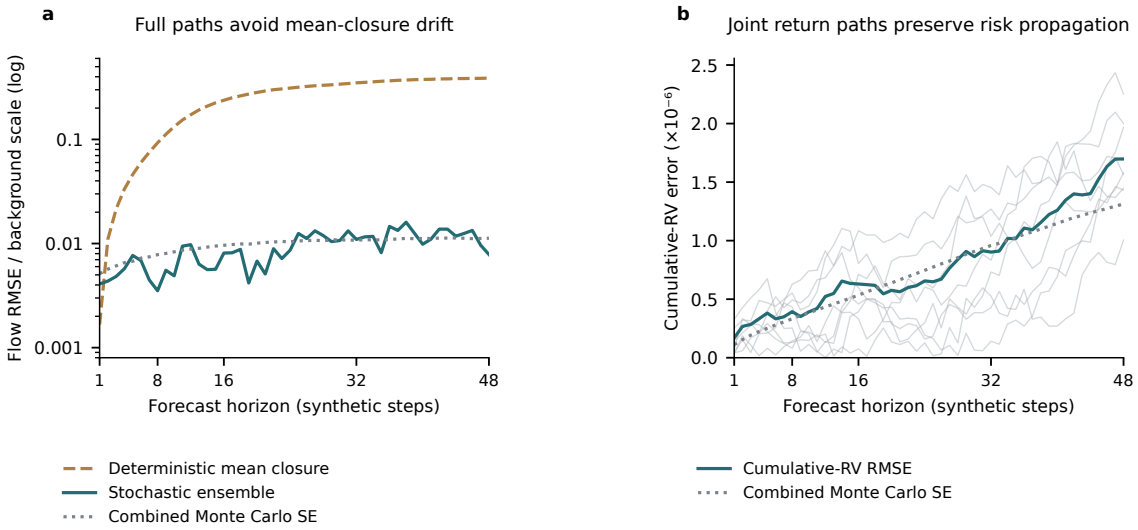


Figure 10. Full-state stochastic propagation and mean-state iteration against independent large-ensemble references. Eight fixed validation states, three pulse conditions and all declared horizons are retained. Monte Carlo agreement evaluates the propagation calculation, separately from latent-state inference and market forecast accuracy. Source: controlled continuation, D01.

9.1 A conditional link from shared information to covariance

For a shared standard Gaussian Z, positive loading $g = 0.15$ and the same fixed prior state, define $t_i(Z) = \tanh[1.5(H_i + gZ)]$. The independent-signal alternative substitutes independent Z_i. Node marginals and mean flow match, while

$$\begin{aligned} &\mathrm{Var}(F \mid S, \text{shared}) - \mathrm{Var}(F \mid S, \text{independent}) \\ &\quad = 2C_0^2 a^2 \sum_{i<j} w_i w_j \,\mathrm{Cov}_Z(t_i(Z), t_j(Z)) \geq 0. \end{aligned} \tag{27}$$

Both functions increase in the same scalar input. With an independent copy Z', their covariance is one-half the expectation of $(t_i(Z) - t_i(Z'))(t_j(Z) - t_j(Z'))$, a nonnegative product. This standard covariance argument connects a shared strategy input directly to Eq. (14). Opposite loadings and changing activity or accommodation are outside this conditional inequality.

Numerical quadrature at all eight frozen validation states gives a one-step flow-variance increment of 24.1–35.5% relative to independent signals; 64- versus 128-node quadrature discrepancies are below 1.82×10^{-11} in absolute variance units. Multiperiod return risk is then evaluated through the actual stochastic network rather than inferred from the one-step sign alone.

9.2 Paired economic-channel experiments

Six arms, including the baseline, generate 768 complete 240-step paths from 128 paired seed units; a further 64 baseline paths supply independent reference training. Table 3 reports all five interventions, with the ten RV/alignment comparisons retained under their frozen multiplicity rule.

Table 3. Controlled economic channels in one fixed network configuration. RV changes are percentages of baseline mean RV; alignment changes are percentage points.

Channel	RV change	Alignment change
Common signal	+3.62%	+3.92
Independent signals, same marginals	−1.78%	−0.52
Positive lagged price feedback	+6.24%	+4.31
Negative price feedback/accommodation	−3.99%	−2.74
Common activity contraction	−6.20%	0 exactly

Independent-signal uncertainty intervals cross zero in the primary changes. The common activity rule reduces participation without altering the direction equation, so its alignment statistic is exactly unchanged. It is not a forced-liquidation or strategic-exit mechanism. The experiment identifies how specified shared inputs and feedback alter risk inside this network configuration; the corresponding causal attribution to real quant institutions would require additional evidence.

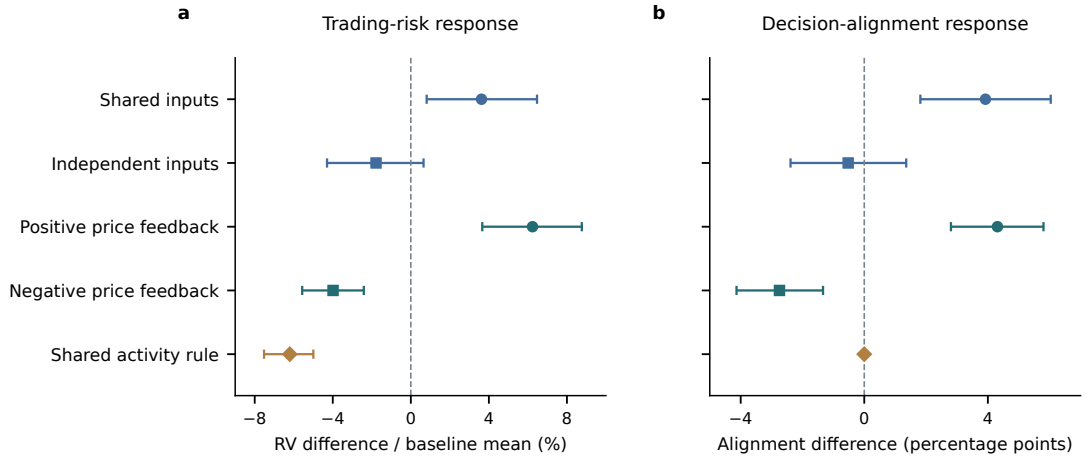


Figure 11. Paired shared-input, independent-input, feedback and activity interventions. The same seed supplies background innovations across arms. Both primary outcomes and every declared intervention are shown; the results concern this fixed source-law configuration.

Evidence record: economic-channel design N01, all paired contrasts and common-input quadrature identity.

10 Observable-state inference and coherent future risk

10.1 From an observable prefix to a distribution of effective states

The inverse task now asks for risk-relevant effective state rather than every edge or participant. A student estimator receives past F, V, K, r; true weights, graphs, node states and future paths are withheld. The supplied 48-unit law and graph/weight distribution are explicit prior information. The first causal particle implementation uses 64 training, 32 selection and 64 new test trajectories. Future-data mutation leaves its predictions unchanged. Static-parameter resampling, however, reduces every tested graph/weight ancestry to one source, yielding zero-width HHI intervals. This implementation check motivates preserving multiple structural possibilities rather than interpreting a single remaining graph as identification.

A conditional simulation ensemble uses the same unchanged network law. Its bank contains 2,048 independent full trajectories, with separate 32-trajectory tuning, 64-trajectory calibration and 128-trajectory evaluation sets. Only 13 observable summaries of the past 32 bars are used to condition the bank. Standardization is fitted on the bank; neighborhood size is chosen among 32, 64 and 128 using tuning-set eight-step QLIKE, selecting 128. Every evaluated case retains 128 distinct source trajectories and their full future joint paths. This is approximate conditional simulation, not an exact Bayesian posterior.

For common nonnegative normalized scenario weights ω_j, the next-step flow moments use total variance:

$$\bar{m}_F = \sum_j \omega_j m_F(S_j), \tag{28}$$

$$\mathrm{Var}(F \mid y) \approx \sum_j \omega_j v_F(S_j) + \sum_j \omega_j [m_F(S_j) - \bar{m}_F]^2.$$

The second term carries uncertainty about hidden state. Table 4 evaluates that state estimate on known-truth simulated units, not on unobserved market participant labels.

Table 4. Conditional simulation state estimates on 128 independent evaluation trajectories. The unconditional comparison uses the same simulation bank. Coverage concerns raw 90% state intervals.

Target	Conditional RMSE	Prior RMSE	Coverage
Fast direction	.17567	.39313	95.31%
Slow direction	.17015	.42501	96.09%
Adaptation memory	.09030	.28313	98.44%
Accommodation	.12295	.15641	88.28%
Weight concentration (HHI)	.00477	.00505	83.59%

HHI intervals have mean width .00977, retaining uncertainty rather than a single repeated ancestor. Their 83.59% coverage is below nominal 90%; source diversity does not establish calibrated structural recovery. The simple F/V fast-direction proxy has RMSE .13519, so the next inferential target is incremental information in slow state, memory, concentration and accommodation. The two inference designs use different evaluation samples and are not a same-sample before/after improvement experiment.

10.2 Generate return and flow moments from common paths

Network RV forecasts average squared returns generated by the nonlinear source law. A local decomposition $r = \alpha + \beta F + \eta$ checks their accounting without assuming residual orthogonality:

$$\begin{aligned}\mathbb{E}r^2 = \alpha^2 + 2\alpha\beta\mathbb{E}F + \beta^2\mathbb{E}F^2 + \mathbb{E}\eta^2 \\ + 2\alpha\mathbb{E}\eta + 2\beta\mathbb{E}(F\eta).\end{aligned} \tag{29}$$

Common weights retain joint flow, accommodation, return and residual paths and preserve covariance positive semidefiniteness. All conditional-ensemble forecasts are positive without clipping. In contrast, fitting cross moments separately had produced 40 non-PSD cases in 1,192 selected rolling fits and 12 nonpositive calibrated risks. The coherent-path construction fixes that numerical inconsistency while leaving forecast accuracy to a separate test.

Expected H-step RV is $\sum_{h=1}^{H}\mathbb{E}(r_{t+h}^2 \mid \mathcal{I}_t)$. The second moment of a cumulative return additionally contains cross-time terms. These two outputs remain distinct throughout the experiments.

10.3 Separate predictive uncertainty from latent-state uncertainty

After freezing a positive point predictor $\widehat{h}(x)$, independent calibration units provide scores $s_i = |\log(y_i/\widehat{h}_i)|$. For nominal coverage $1-\alpha$,

$$\begin{aligned}k = \lceil (n+1)(1-\alpha) \rceil, \quad q_\alpha = s_{(k)}, \\ C(x) = [\widehat{h}(x)e^{-q_\alpha}, \widehat{h}(x)e^{q_\alpha}].\end{aligned} \tag{30}$$

For $k = n+1$ the order statistic is defined as infinity. With $n = 64$ and $\alpha = .1$, $k = 59$. This standard split-conformal construction[14] gives marginal coverage under exchangeability of independent calibration and evaluation units, conditional on earlier fitting/selection. It is not state-conditional coverage or a guarantee for dependent market dates. Risk-interval calibration does not calibrate HHI intervals.

A frozen 48-step particle-model evaluation uses 64 fresh calibration and 128 fresh evaluation trajectories. Raw intervals cover 118/128 in that new evaluation; calibration covers 121/128 with a larger mean width and slightly worse Winkler score. Its point QLIKE is .03594, compared with .03562 for the unconditioned source-law prior, with paired difference interval $[-.003365, .003888]$. Figure 12 retains this earlier particle-model test, separately from the conditional-ensemble comparison below.

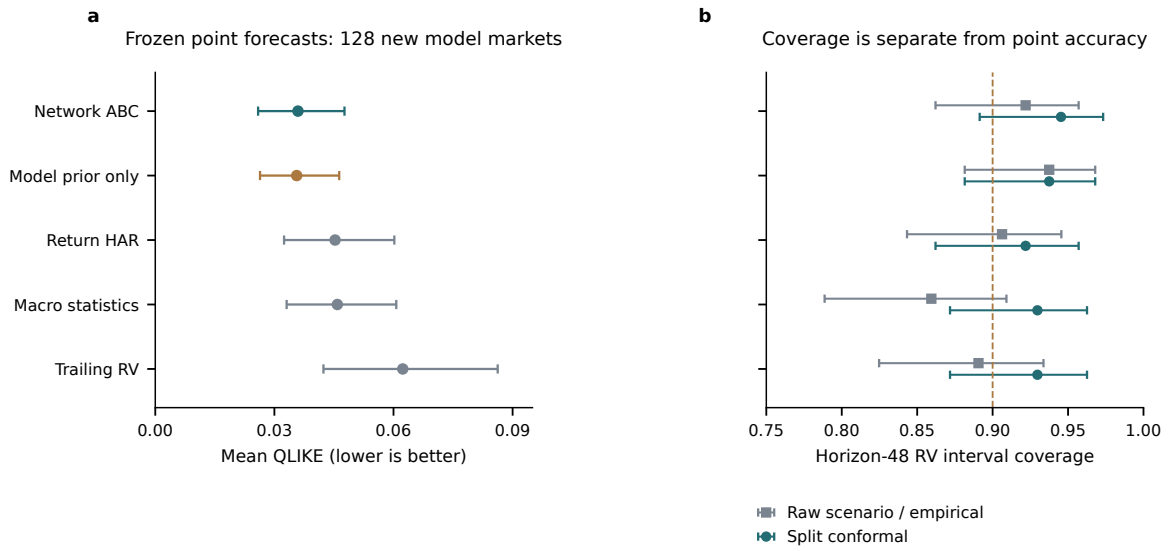


Figure 12. Observable-only inference and joint risk in the particle-model experiment. Source-law priors, test-state validation, interval width and calibration are displayed separately. This figure uses the first inference design and its frozen new calibration/evaluation sample; Table 5 uses a distinct conditional-ensemble design.

10.4 Risk comparison with equal simulation labels and macro features

The conditional ensemble is compared with a macro Gamma model using the identical 13 observable features and the same bank of future-risk labels. Fitted controls use the same bank and future-risk labels; the unconditional prior averages that bank, while trailing RV is a prefix-only baseline. Every method receives the same 64 interval-calibration labels. Eight steps is the fixed primary horizon, with one and 48 retained as secondary horizons.

Table 5. Primary eight-step risk on 128 independent controlled evaluation trajectories. QLIKE is smaller for better point forecasts. Coverage and width follow the shared 64-unit interval-calibration budget.

Method	QLIKE	Coverage	Mean width
Conditional network ensemble	.13412	95.31%	3.8×10^{-5}
Same-feature macro Gamma	.13329	94.53%	4.0×10^{-5}
Unconditional network prior	.13688	95.31%	4.1×10^{-5}
Return-HAR	.13570	95.31%	4.0×10^{-5}
Trailing RV	.19185	94.53%	4.9×10^{-5}

The paired QLIKE difference between the ensemble and Gamma is +.000837 (95% interval $[-.004832, .006098]$). Relative to the unconditional prior, the difference is −.002756 (interval $[-.015453, .008894]$). Incremental risk value over those two primary references is not established. Improvement over trailing RV survives the frozen four-comparison Holm rule, but that weaker reference does not replace the equal-feature and prior controls. At 48 steps, point scores .04304 and .04438 for ensemble and Gamma are reported as secondary outcomes, not a replacement primary horizon.

The full generative law, model age, unobserved initialization, offline state/risk labels, retained history and calibration are counted as information costs. A 32-bar deployed interface does not by itself establish savings over a statistical method using a longer observed history. The conditional-ensemble experiment establishes a working macro-to-state-to-risk calculation and a direct test of its added predictive value under disclosed simulation information.

Evidence record: observable inference I01, joint risk R01, calibration R02; conditional-ensemble state/risk scores and independent numerical recalculation.

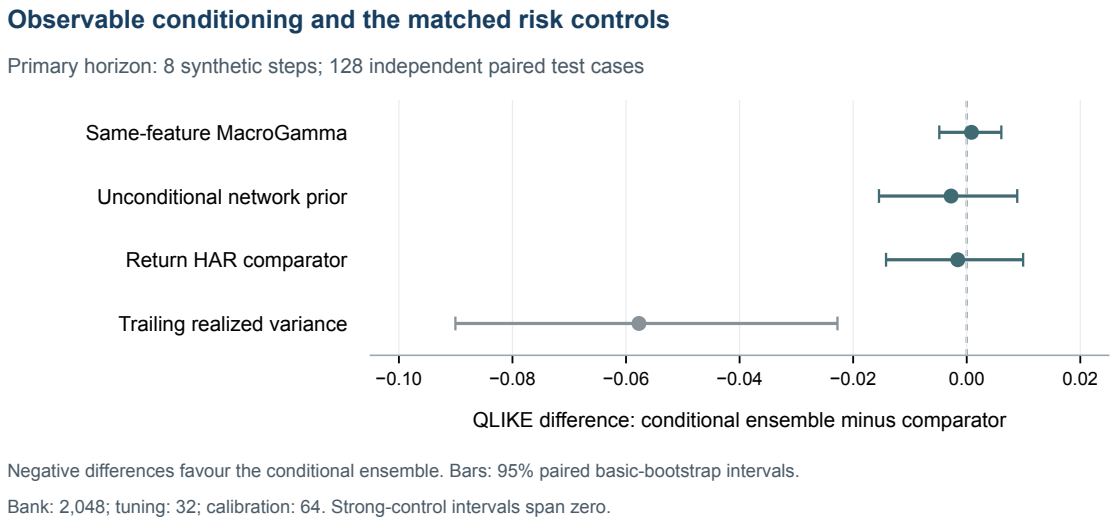


Figure 13. Primary eight-step conditional-ensemble risk contrasts: QLIKE difference from each prespecified reference, with 95% paired basic-bootstrap intervals; negative values favor the ensemble. The design uses 2,048 bank, 32 tuning, 64 calibration and 128 evaluation trajectories. The same-feature Gamma comparison shares observable summaries and simulated risk labels; the known generating law and offline labels remain disclosed costs. The three stronger-reference intervals cross zero. Trailing RV is a weaker comparator. Source: conditional-ensemble risk contrasts.

11 Connecting the network clock to real market observations

11.1 What the unchanged source law does not yet map

A decision unit, an execution and a calendar-time bar are different objects. A common-sign parent can generate multiple exchange executions; a five-minute bar can contain many internal decisions. Financial transaction models treat arrival times and price changes jointly[15], providing a precedent for a separate observation clock. Here the original network continues to supply directional interaction, while an arrival/fragmentation layer is the proposed route to market observations.

A descriptive transport check compares all 663 previously examined Bitcoin days, partitioned into 1,989 blocks of 96 five-minute bars, with 64 new 96-step source-network blocks. Table 6 summarizes scale-independent discrepancies under the tentative direct step/bar map.

Table 6. Uncalibrated direct-map observation comparison. Means concern the stated block summaries; three blocks share each market date and are not independent experimental replications.

Block statistic	Source network	Bitcoin
Lag-one correlation of pressure F/V	.709	.142
Coefficient of variation of volume	.150	.928
Coefficient of variation of count	.123	.626

The direct map carries excessive directional persistence and too little activity variation. Raw exchange counts also lack a calibrated relation to 48 effective active units, and 31 real return bars exceed the unchanged source return cap. The model was therefore not evaluated as an independent new-market predictor. These diagnostics localize the observation problem rather than treating normalized features as a substitute for units, clocks and response calibration.

11.2 A minimal next model and its acceptance criterion

The proposed extension separates an internal decision clock, child executions and calendar aggregation. A common activity state can govern decision-event intensity and child-execution arrivals; the retained fast/slow network evolves in event time. A size/fragmentation rule then generates observed K, V, F, and a jointly simulated response/accommodation layer produces returns. This is a model proposal, not a completed calibrated market mechanism.

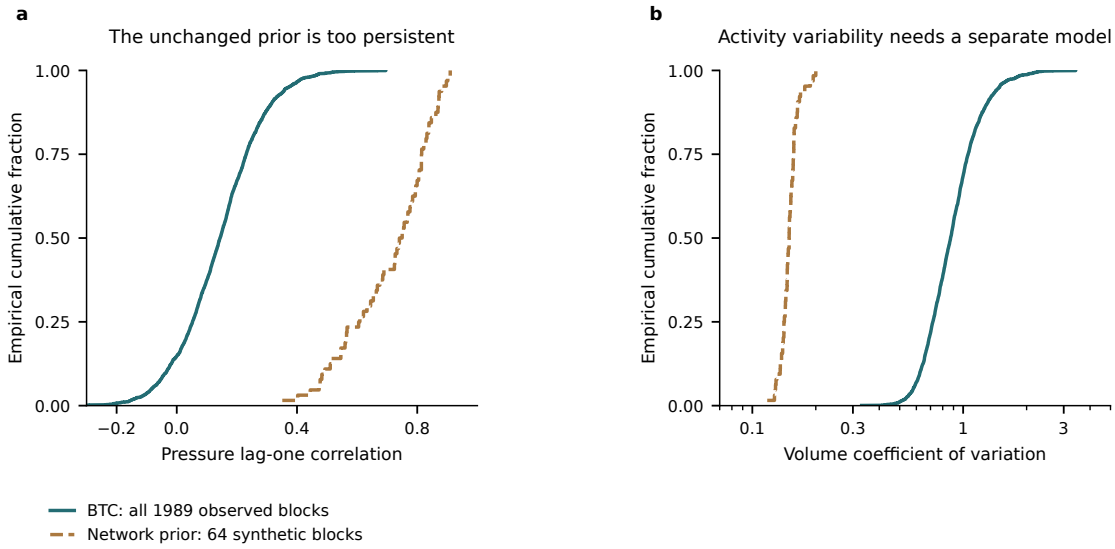


Figure 14. Source-network and Bitcoin observation checks under a tentative direct map. All 663 examined days and 64 new source-network blocks are used. Pressure persistence, activity variation, execution units and return support diagnose the observation layer before an independent market test is opened. Binance Vision-derived statistics retain source attribution.

Calibration should jointly confront pressure correlations, volume/count variation, size concentration and flow/return moments at 5-, 15- and 60-minute scales. A time multiplier matching one correlation is insufficient. Direct F/V can serve as a fast-direction measurement while the network is tested for useful slow-state and accommodation information. If only a range of weight/concentration states is identified, that range should be propagated to risk rather than replaced by a unique trader count.

After the observation map passes, the candidate, history and calibration budgets, accuracy objective and independent dates/assets can be frozen. The practical criterion is lower total information cost at comparable reliability, or better risk accuracy with matched information. Any additional liquidity field must have an explicit identification role and be supplied to every comparator. The current studies identify these tests and provide their implementation basis; an independent real-market confirmation remains a subsequent stage.

Evidence record: market observation comparison V01 and the observation-layer research specification.

12 Historical next-day statistical forecasting

12.1 How physics and forecasting are connected

The physics motivates observable activity, joint price–volume statistics, and extreme-concentration measures. Market tests assess the fixed interface and reveal temporal dependence. The forecasting application directly uses these observables and multiscale lags, without true network or cluster labels. The heterogeneous autoregressive (HAR) representation follows the multiscale approach of Corsi[8]. It is an empirical forecasting specification using mechanism-motivated observables and temporal lags, with forecast accuracy evaluated against named statistical references.

After the final five-minute bar of day d closes, the model forecasts the next UTC day:

$$RV_d = \sum_{j=1}^{288} r_{d,j}^2, \quad z_{d+1} = \log RV_{d+1}, \tag{31}$$

$$\widehat{z}_{d+1} = \bar{z}_{\text{fit}} + \widehat{w}^\top \widetilde{x}_d,$$

$$\widehat{w} = \arg\min_w \sum_{d \in \text{fit}} (z_{d+1} - \bar{z}_{\text{fit}} - w^\top \widetilde{x}_d)^2 + \alpha_R \|w\|^2. \tag{32}$$

Standardization uses the fitting period only. The ridge penalty α_R differs from a tail exponent. The displayed $100\sqrt{RV}$ measures intraday volatility intensity, without specifying the direction of the close-to-close return.

Table 7. The full model's ten inputs, all available at the forecast origin.

Group	Definition	Count
HAR	$\log RV_d$; log mean RV over the past 7 days; log mean RV over the past 28 days	3
Activity	Log total daily BTC volume; log ratio of daily volume to its past-7-day mean	2
Coupling	Bar-level Spearman$(\lvert r\rvert, v)$; $\overline{r_c^2 v_c}/(\overline{r_c^2}\,\bar{v})$; absolute summed signed-taker volume divided by daily volume	3
Tail	$\log[288\sum r_j^4/(\sum r_j^2)^2]$; $\max r_j^2/\sum r_j^2$	2

r_c, v_c are demeaned within a day. Signed activity derives from venue taker fields, not full order-book OFI or market-wide buy–sell imbalance. Tail inputs measure finite-sample shape and concentration, not power-law exponents. The 1/7/28-day horizons adapt the setup to a 24/7 market.

The original protocol used 337 eligible training targets in 2025 and 90 validation targets in 2026 Q1. After selection, one refit used all 427 targets. Parameters remained fixed across 183 evaluation dates from April 1 to September 30, 2026. Observed test-day data can become later lagged inputs, but do not enter refitting or renewed selection. Figure 15 retains every test date.

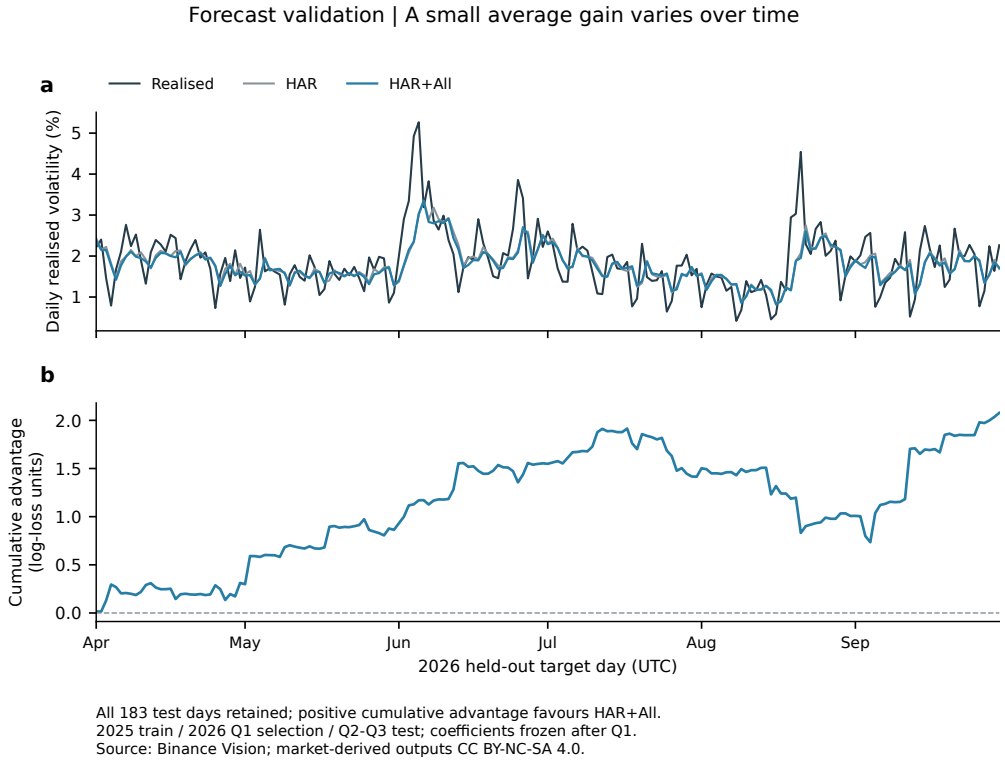


Figure 15. Forecasts and realizations on all 183 historical evaluation dates. Volatility is $100\sqrt{RV}$. The lower panel accumulates forecast-loss advantage over HAR. Binance Vision, CC BY-NC-SA 4.0. [P01]

12.2 Overall skill and incremental value are separate

For D evaluation days, the primary loss and relative skill are

$$L_{\log} = D^{-1}\sum_d (\log RV_d - \widehat{z}_d)^2,$$
$$\text{Skill}_B = 100(1 - L_{\text{model}}/L_B). \tag{33}$$

Table 8 and Fig. 16 place simple and strong references together. The full model has positive historical skill over a constant, persistence, and EWMA. HAR alone already improves over persistence by 29.61%. The additional 2.263% point gain over HAR has an interval crossing zero, with Holm-adjusted $p = 0.3948$ in the original primary comparison. This contrast measures the combined addition of return-shape and trading-information features.

Table 8. Original next-day log-RV evaluation: 183 BTCUSDT dates.

Reference/model	log-MSE	All loss reduction (%)	Approx. 95% CI (%)
Fit-period mean	0.793999	38.26	[24.65, 48.97]
Persistence	0.712561	31.20	[21.76, 38.62]
EWMA	0.606913	19.23	[13.22, 24.99]
HAR reference	0.501565	2.26	[-0.11, 4.56]
HAR+All	0.490213	—	—

All denotes the original HAR+All. Intervals use 4,999 paired circular seven-day blocks, unadjusted, in a post-analysis overall-skill reassessment. Source: predictive_utility_v1.

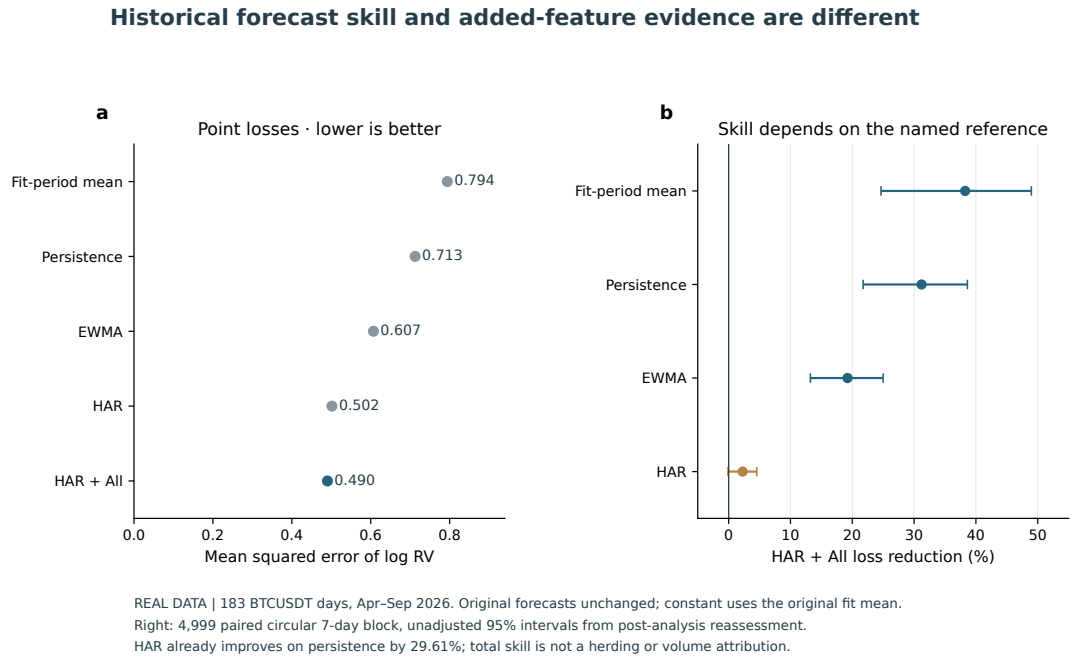


Figure 16. Historical log-MSE and full-model loss reductions against named references. Intervals use 4,999 paired circular seven-day-block resamples. The constant-reference comparison belongs to the post-analysis reassessment; the original HAR comparison is reported in the text. [J02]

The output is a next-day volatility estimate paired with a date-by-date error record. Subsequent asset and time tests in Appendix D examine the transfer of the added features. Prospective calibration and decision-specific losses would be needed to assess operational risk use.

12.3 Mechanism-imposed risk under a matched information budget

The statistical feature benchmark above asks whether selected observables improve a HAR forecast. A separate mechanism test asks whether a pressure-state restriction adds information while preserving the same raw inputs and total-history budget. Both questions are retained, with their own protocols and scores.

The rolling mechanism study uses the same past price, volume, count and signed-flow fields as its macro-HAR reference. A reduced pressure model forecasts future flow moments and passes them through the response/risk map. At a 240-day total-history budget, its QLIKE is .260805; a same-map pressure AR control gives .260792, return-HAR .228191, and macro-HAR about .2236. The near identity with the pressure control means the tested network-motivated reduction supplies

no demonstrated increment at this budget. The historical dates have been examined in prior project rounds and are development evidence, not a newly untouched test.

Further joint-flow-moment correction is evaluated at 30, 60, 120 and 240 days rather than selecting one favorable budget. At 240 days its QLIKE is .2749, versus .2236 for macro-HAR. Separately estimating residual cross moments produced incompatible covariance fits; the common-path construction in Section 10 addresses that calculation issue without retroactively changing these market scores.

A signed-flow/response bridge also generates sequential $\Delta = -9.562$, compared with -8.181 observed, while its selected additional forecast channel is Off. Its matched-HAR QLIKE .183724 and selected shared hybrid .186232 belong to that different fixed protocol. Thus explaining a negative compatibility statistic and improving a next-day forecast are separately evaluated achievements. Scores from distinct protocols are not interpreted as a chronological improvement series.

Evidence record: matched-budget collective-dynamics scores, joint-flow W3 comparisons and the chronological signed-flow bridge benchmark.

13 Portfolio and strategy risk diagnostics

13.1 How much sample risk can common movement conceal?

The US-equity study contains nine sector ETFs, SPY and QQQ references, and 46,530 daily rows through October 23, 2026. Financial association networks have established precedents[9]; our sector graph represents observed co-movement, not trader interactions. The original design uses 2010–2017 for training, 2018–2019 for selection, 2020–2025 for historical evaluation, and 2026 for confirmation.

For within-window return covariance $\widehat{\Sigma}$ and weights w,

$$\widehat{v} = w^\top \widehat{\Sigma} w = \underbrace{\sum_i w_i^2 \widehat{\sigma}_i^2}_{v_{\text{self}}} + \underbrace{2 \sum_{i<j} w_i w_j \widehat{\sigma}_{ij}}_{v_{\text{collective}}} . \tag{34}$$

Annualized volatility is $100\sqrt{252\widehat{v}}$. For nine equally weighted sectors in the 63-day window ending March 31, 2020, the full estimate is 57.73%. Retaining only diagonal terms gives 20.62%, a ratio of 2.80. Collective terms account for 87.24% of full estimated variance. Both estimates use the same returns, weights and 63-day window; their variance difference isolates the sample covariance contribution. Figure 17 retains the longer rolling history.

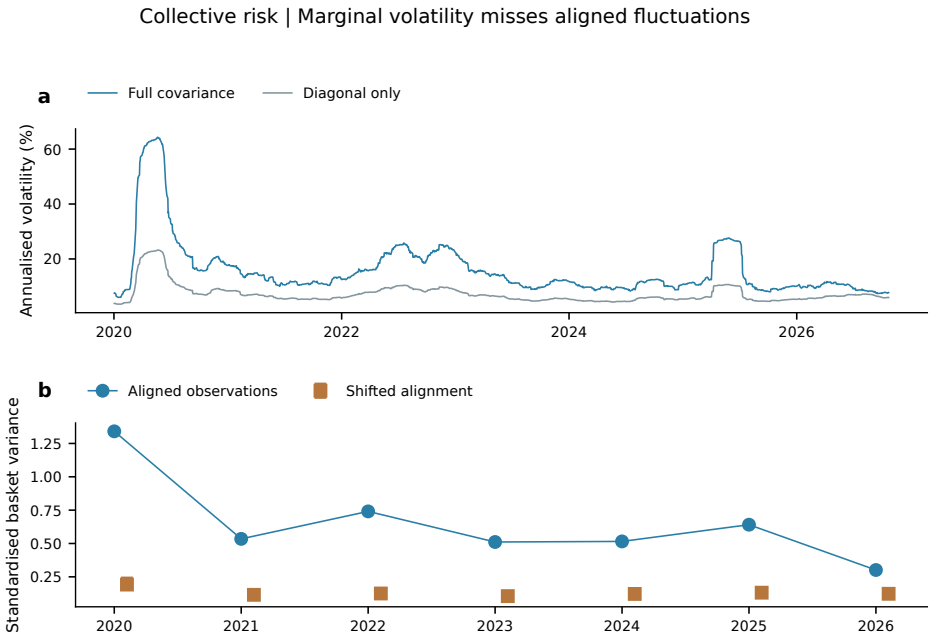
The existing tool lets users change dates, 63/126-day windows, and holdings. It also evaluates $\Sigma_\eta = (1-\eta)\,\mathrm{diag}(\widehat{\Sigma}) + \eta\widehat{\Sigma}$, with $0 \le \eta \le 1$. The interpolation preserves positive semidefiniteness and gives a conditional covariance scenario for the selected holdings. The interface exposes how estimated common movement changes the same portfolio's risk.

13.2 Strategy variants are not independent risk sources

The six fixed proxies use momentum at 63/126 days, reversal at 5/21 days, and low volatility at 21/63 days. Dependence is assessed primarily through daily net P&L increments. The low-volatility pair has daily correlations 0.921/0.795 in the historical/2026 periods, versus cumulative-curve correlations around 0.982/0.985. Cumulative curves therefore cannot replace daily-risk analysis. Figure 18 separates shared exposures and cross-asset terms.

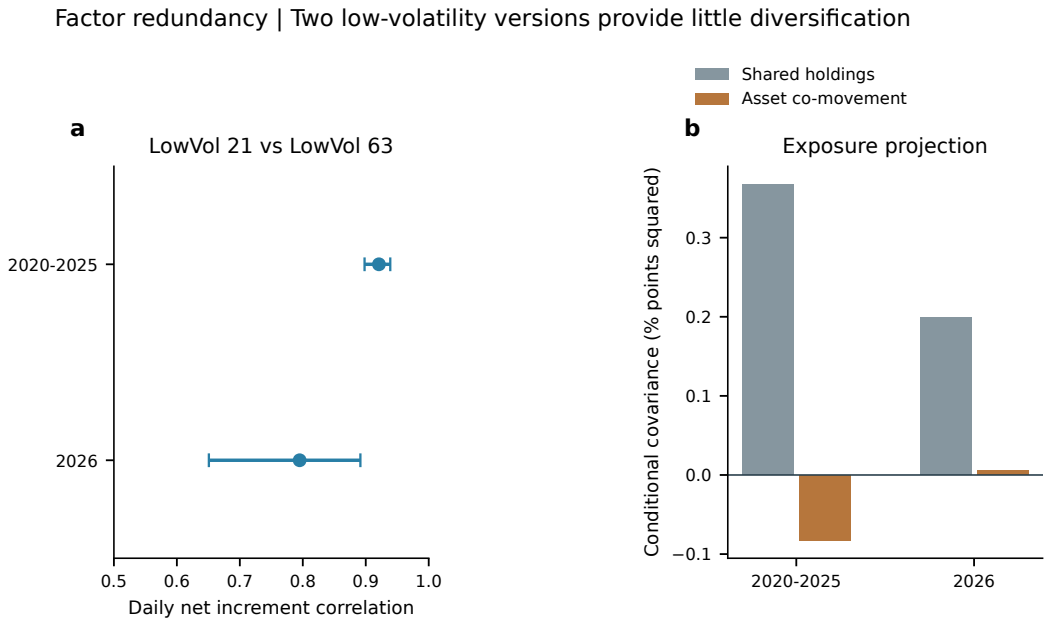
The default screening rule uses only 2011–2017 daily net increments and turnover. It ranks candidates by turnover and flags positive correlations ≥ 0.80. Six candidates become five representatives, reducing their count by 16.7%. Evaluation-period statistical reconstruction of the flagged strategy gives $R^2 = 83.9\%/64.1\%$. The rule was proposed after evaluation data had been inspected and remains exploratory. Reconstruction R^2 is neither retained alpha nor an executable hedge return.

Table 9 shows higher combined turnover after removing one candidate, together with higher 2026 volatility. The useful workflow flags candidates, then checks combined holdings, costs, and exposures before deciding what to retain. The interface makes all 63 nonempty subsets available for comparing those consequences under a fixed evaluation period.



a: Rolling 63-session estimates, descriptive. b: Per-asset trailing-scale standardisation.
499 within-year shifts; bars show central 95% of surrogates, not a sampling CI or causal test.
Data: Yahoo Finance daily ETF panel. Fixed research rules; no live trading claim.

Figure 17. Sector-portfolio risk decomposition and conditional shifts of cross-asset date alignment. Numerical risk uses the full covariance matrix, independently of the plotted edge threshold. Shift ranges describe the conditional alignment comparison. The fixed 2020 case is in risk_lab_v1/risk_snapshots.csv. [U02]



a: 20-session paired blocks, 4,999 resamples; post-analysis descriptive pair diagnostic.
b: Average past-covariance projection, no causal partition. Equal-risk effective count: 1.04 / 1.11.
Data: Yahoo Finance daily ETF panel. Fixed research rules; no live trading claim.

Figure 18. Daily net P&L dependence for the two low-volatility strategy variants, with cumulative-curve comparisons and exposure-based risk decomposition. Intervals use date blocks. For equal-risk strategies with daily P&L correlation c_{12}, $2/(1+c_{12})$ is a covariance-equivalent count. [U05]

Table 9. Default screening from six strategies to five representatives.

Period	Cumulative net P&L (%)	Annual P&L volatility (%)	Daily turnover (%)
2020–2025	-21.68→-20.77	4.35→4.33	7.46→8.67
2026 confirmation	-6.44→-5.57	3.84→3.99	8.03→9.05

Fixed reference capital with additive P&L; 5 bp one-way transaction cost and assumed 100 bp annual borrowing cost. Screening is post-analysis exploration, not a new blind test. Source: risk_lab_v1/subset_metrics.csv.

13.3 How the mechanism enters the tool

With additional assumptions of common loading v, independent noise covariance D, and fixed strategy exposures W,

$$\Sigma_Z = Kvv^\top + D, \quad K = q(\lambda/N)^2 \mathbb{E}S_2,$$
$$\Sigma_{\text{strategy}} = W\Sigma_Z W^\top. \tag{35}$$

This conditional factor projection connects a coordination-generated common mode to the covariance used by the risk tool. Empirical covariance can also reflect common news, holdings, and heteroskedasticity. The projection describes fixed exposures; daily strategy P&L is evaluated separately with its actual time-varying holdings.

The existing single-file tool has four modules: omitted risk, factor screening, allocation comparison, and physical mechanisms. Its mechanism module varies ρ, q, redraws graphs, and simulates actions while displaying cluster moments and model risk. The common-loading scenario is an illustrative model calculation; sector risk uses the observed return covariance. Tool settings, source metrics, and implementation checks accompany the interface.

Source tables and methods: `web/advanced/index.html`.

14 Discussion

14.1 The useful inferential object is effective statistical structure

The finite reference model supplies exact moments, parameter recovery under known scales and a structural identification boundary. The weighted extension retains its core covariance mechanism while distinguishing effective trading quantity from participant head count. Microscopic execution checks then show which parts of that description public macro observations constrain and where size calibration is needed. High count dispersion or a large energy-amplification ratio does not alone select a unique coordination mechanism.

The dynamic studies extend the same object rather than replacing it with a disconnected predictor. Fast/slow interdependence, inertia, adaptation and accommodation determine future flow distributions. Exact one-step formulas and verified stochastic propagation connect those mechanisms to risk. Shared inputs provide a concrete source of covariance; paired positive and negative feedback experiments show how the resulting price-risk effect depends on the economic response channel.

14.2 Prediction is a separate task with a complete information cost

Observable-only inference and common-path risk make the macro-to-risk route executable without revealing the target's hidden state or future. Distinct tests evaluate structural coverage, proper forecast losses, risk-interval calibration and market observation fit. The present evidence supports the conditional calculation and its diagnostic use; neither reliable market structural recovery nor incremental risk accuracy or information saving has been established.

The direct F/V proxy already captures much of the tested fast-direction information. A focused next inverse task is whether slow state, effective concentration or accommodation adds future-flow information beyond that proxy and equally supplied statistical references. Known simulator laws, offline labels and calibration data belong in the cost of that inference. Risk calibration and structural identification remain separate: widening a risk interval cannot certify recovered weights.

14.3 Applications and research directions

The portfolio and strategy tools expose shared-risk contributions that are directly computable from holdings and return observations. Historical replay links observed large executions, size baselines and conditional risk estimates along a common clock. These are useful explanation and exposure-audit interfaces; their labels distinguish observations, estimates and selected replay outcomes. They do not attribute market risk to specific quant institutions.

The next empirical step is an event/arrival and fragmentation observation layer for the retained network, calibrated jointly across time scales and with uncertain price response. It addresses measured activity and persistence discrepancies before opening a new market evaluation. An operational risk application would additionally specify its horizon, acceptable error, action and decision loss.

Quantum models and quantum machine learning remain possible extensions for collective dynamics and inference. A focused first comparison could reuse the known-truth effective-state task and compare observation needs, error and computational cost with moment inversion and classical simulation inference. Identical classical observation laws remain identical under a common data encoding, so the target must be identifiable or use genuinely different observations. The role of data and task-specific nature of a learning advantage[12] determine whether and when such an extension is appropriate.

15 Conclusion

Aggregate trading observations support a precise research route from collective covariance to effective state and risk. Exact finite-network moments and observational equivalence establish what a tractable reference can identify. Effective weights, execution-unit checks and temporal covariance extend the interpretation to large decisions and fragmented trading. The retained interdependent decision network supplies exact local flow moments, verified future-distribution propagation and controlled common-signal/feedback effects. Observable-prefix conditioning, coherent joint risk and separate calibration make that route computationally testable, while matched-information comparisons and market-clock diagnostics locate the remaining requirements for reliable deployment. The contribution is a continuous, reproducible mechanism-and-inference framework with explicit tests of both its explanatory content and its predictive value.

A Model context: adaptation, bubbles and crashes

The model families in Table 10 address complementary mechanisms. Cont–Bouchaud links random coordination and shared cluster actions to aggregate fluctuations[1]. Our finite implementation specifies both volume observation and unit-order execution. Lux–Marchesi adds switching between opinions and trader types, producing endogenous feedback and temporal structure[2]. The persistent-state experiment in Section 6 isolates temporal activity and response while retaining exogenous state dynamics.

Johansen–Ledoit–Sornette (JLS) relates reinforcing imitation to conditional bubble growth and stochastic crash risk[3]. Its simplified setup is

$$dp/p = \mu(t)\,dt - \kappa\,dj, \quad \mathbb{E}[dj] = h(t)\,dt, \quad \mu(t) = \kappa h(t). \tag{36}$$

The hazard $h(t)$ links growth conditional on survival to compensation for jump risk. The study discusses pre-crash patterns for Wall Street in 1929, 1962 and 1987, and Hong Kong in 1997. These published retrospective cases illustrate the bubble mechanism. The endpoint of conditional bubble growth and the random crash time are distinct model quantities.

B Complete tail and scaling objects

Cluster number density, node-weighted cluster size, and return distributions are different observables. Figures 19 and 20 retain cluster evidence and separate population size from temporal aggregation.

Table 10. Mechanisms represented by three financial market models.

Model	Mechanism	Connection to the present analysis
Cont–Bouchaud[1]	Shared actions in random groups	Finite ER model, price–volume moments, inversion and compatibility tests
Lux–Marchesi[2]	Opinion and trader-type switching with profit feedback	Adaptive comparison for the temporal-state experiments
JLS[3]	Reinforcing imitation, hazard and conditional growth	Literature context for bubbles, crashes and criticality

Independent-model second-order aggregation differs from market memory.

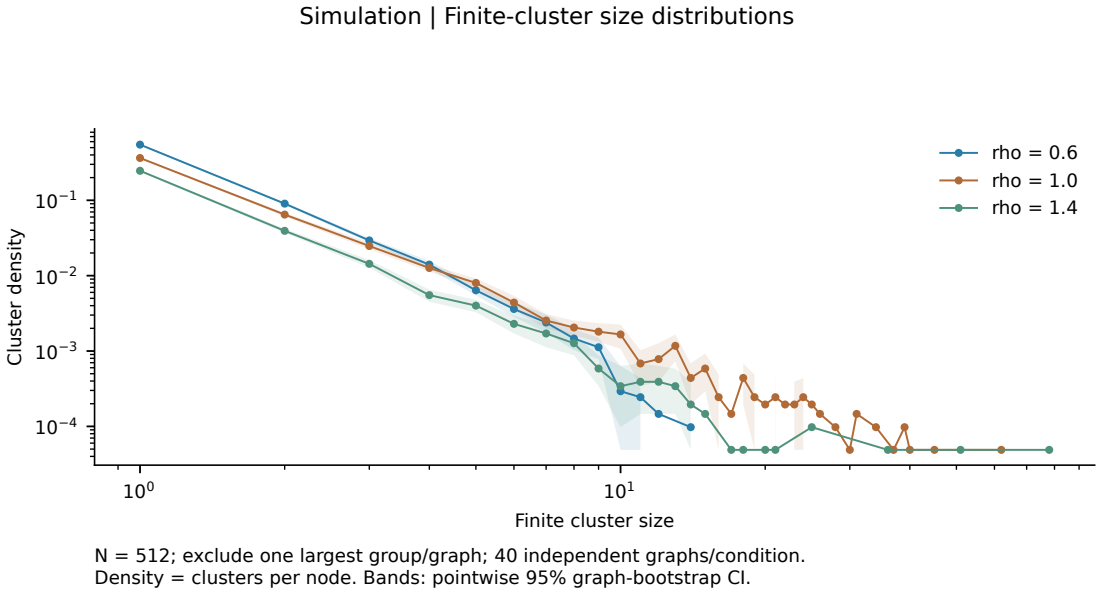


Figure 19. Finite-cluster number density at the specified population size. Pointwise bands resample graphs. Uniform-node sampling would weight each cluster by its size; raw cluster sizes and all population-size source tables are retained. [F04]

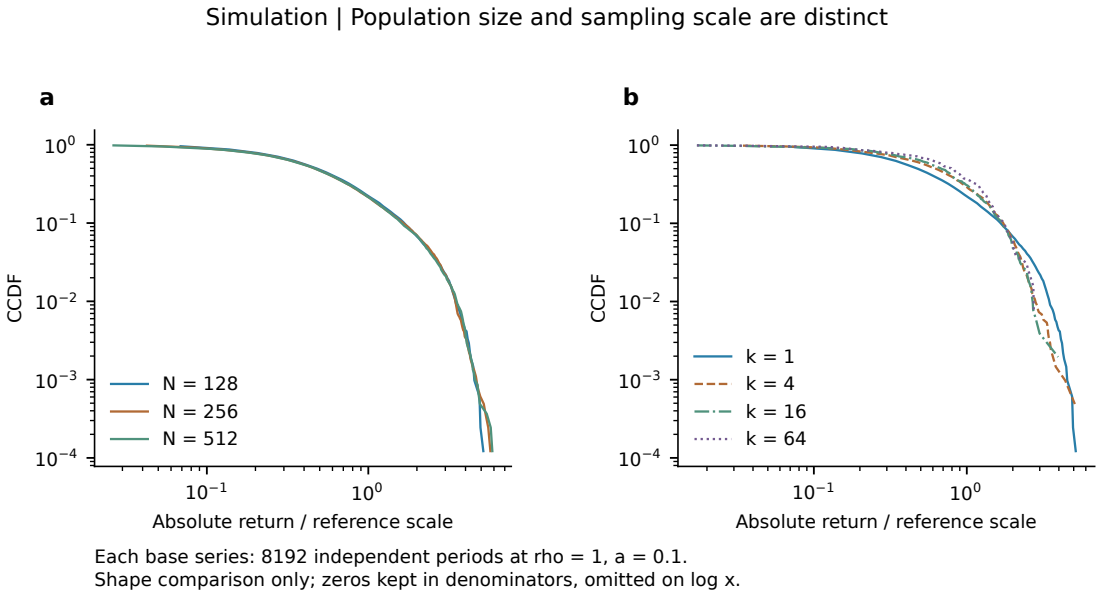
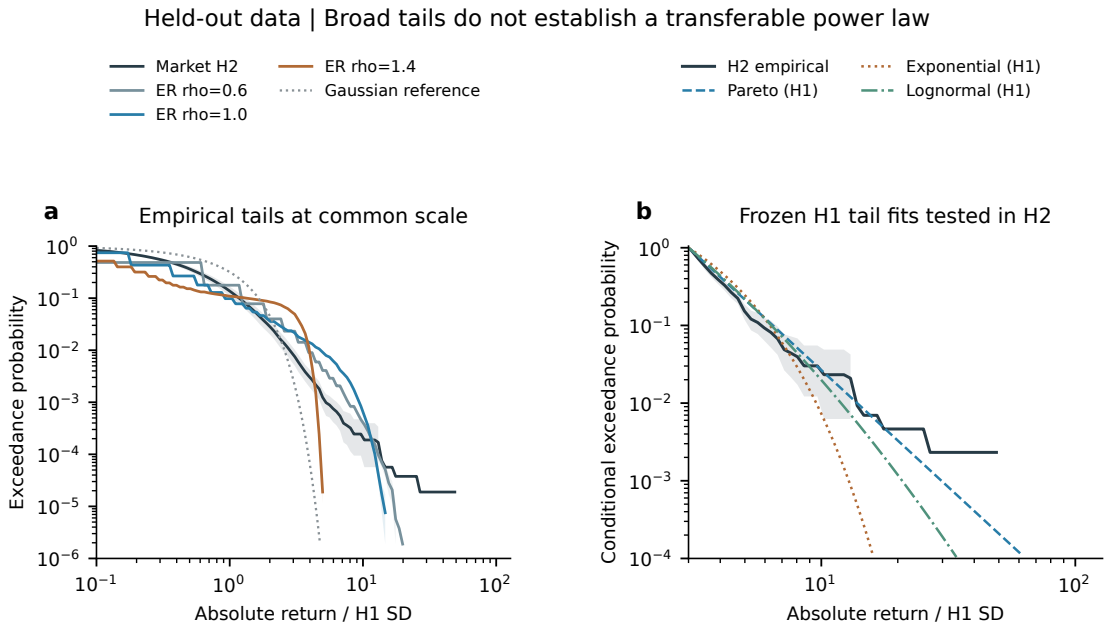


Figure 20. Separate comparisons of population size and temporal aggregation. Return standardization permits shape comparison; temporal blocks are nonoverlapping. The source tables retain all conditions. [F06]



H1 return SD fixes units; tail threshold: 3 H1 SD; H2 exceedances: 431.
Market bands: 7-day blocks; model bands: independent-path bootstrap.
Source: Binance Vision; market-derived figures CC BY-NC-SA 4.0.

Figure 21. H1-scaled H2 market return CCDFs, conditional tails and continuous alternatives. Zero exceedances remain in the source data and are omitted from logarithmic display. Binance Vision, CC BY-NC-SA 4.0. [R02]

Figure 21 adds the calibration-scaled market/model tail comparison to the joint-observation and memory diagnostics. All 28 market CSVs and original reproduction records are retained. Source maps distinguish model seeds from observable inputs.

C Exact checks and dependence controls

Theory files contain proofs of Eqs. (2)–(3), monotone identification, and chain/star and mixture equivalence. Rational enumeration and exact certificates check implementations. The original 480 theory checks and ten-study refit records are included. Small-graph enumeration and rational certificates provide independent computational checks of the stated identities. The marginal-preserving permutation comparison D03 isolates same-period coupling: shuffling volume preserves both empirical marginal distributions while changing their joint dependence.

D Supplementary forecasts and external evaluations

For a positive realized variance y and positive prediction μ, the quasi-likelihood loss (QLIKE) is

$$\ell_{\mathrm{QLIKE}}(y, \mu) = y/\mu - \log(y/\mu) - 1. \tag{37}$$

QLIKE is a secondary loss in the original log-RV benchmark and the primary loss in the subsequent state-forecast study. The latter's Gamma specifications use a log-linked positive mean and separate penalties on return and trading-information channels. Off denotes the matched return-only model with the trading-information channel fixed to zero. Feature and fitting definitions are in `docs/AB/state_forecast_v2_feature_spec.md`.

Table 11 retains supplementary losses. Monthly point gains over HAR are 2.90%, 5.41%, 4.99%, −1.12%, −1.71%, and 5.15%. A post-analysis comparison against HAR+Tail gives a 1.535% point gain, again with an interval crossing zero. The final evaluation contains eight high-volatility events, giving a small event-specific assessment sample.

The subsequent state study adds an external ETH evaluation and 25 new dates for each of BTC and ETH. BTC's new-time primary QLIKE gain is 2.476%, with approximate 95%

Table 11. Supplementary forecast losses, without replacing the primary metric.

Model	QLIKE	Volatility MAE (percentage points)
HAR	0.234214	0.440987
HAR+All	0.233933	0.433735

Volatility is $100\sqrt{RV}$. Exponentiating a log prediction need not yield a conditional arithmetic mean.

interval $[-0.407, 4.815]\%$. Validation selects Off for the ETH trading channel. Complex Gamma specifications do not consistently beat simple HAR, as retained in Fig. 22. The BTC and ETH new-time evaluations cover the same 25 calendar dates.

On the same short BTC period, supplementary log-MSE improves by 3.170% against matched return-only Gamma. Its approximate 95% interval is $[1.236, 4.733]\%$, with approximate Holm $p = 0.0276$ across that three-test supplementary family. This contrast assesses the trading-information channel within the Gamma specification. The primary assessment uses QLIKE and retains simple HAR among the references. Twenty-five dates provide only about 3.6 seven-day blocks, requiring a strong dependence approximation.

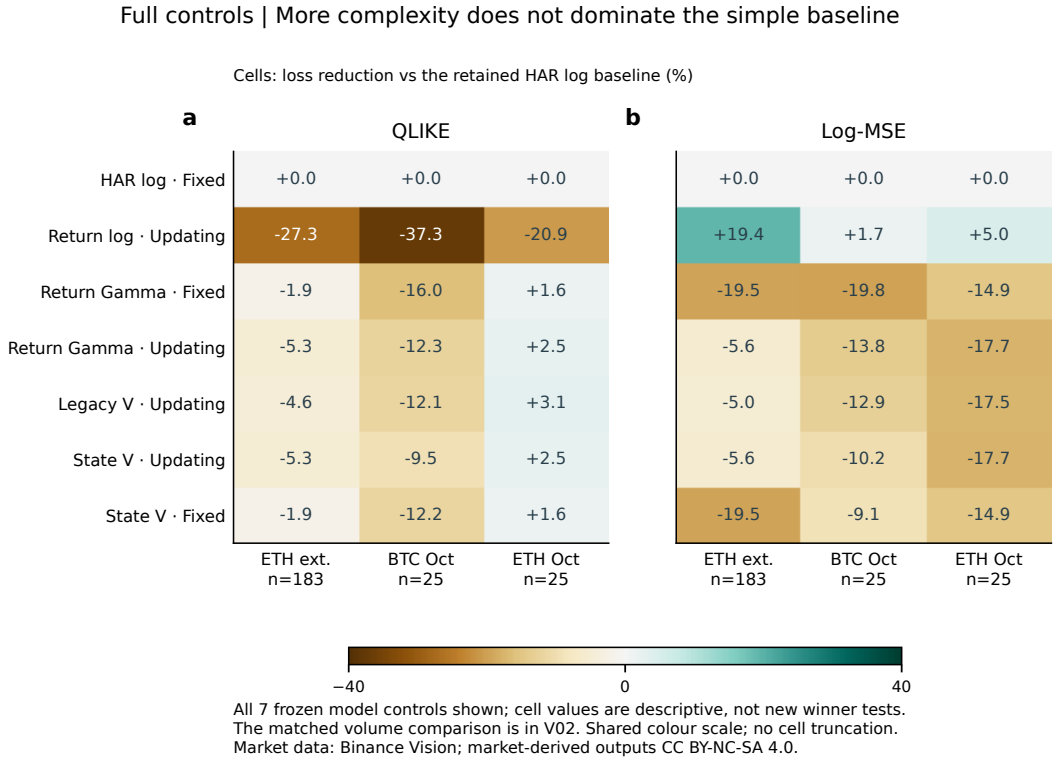


Figure 22. State/price–volume forecasts with simple HAR and matched controls. Previously inspected BTC periods are development information; external-asset and new-time evaluations are reported separately. Off denotes a validation-selected return-only fallback. Binance Vision, CC BY-NC-SA 4.0. [V03]

Covariance-matched tail tests hold information and covariance fixed, comparing nine methods on 38,070 rows from nine external stocks. Historical tail improvement is 0.557%, with Holm $p = 0.0772$; the 2026 point change is -1.797%. There are 32 historical co-crash events and zero in 2026, restricting the latter evaluation to false alarms. Figure 23 and the complete controls do not establish stable market increment. The external-stock evaluation tests transfer across assets over an already elapsed historical period.

E Stopped capacity extension and volume trade-offs

Six strategies, nine ETFs, 190 monthly snapshots, four allocations, and three depth levels produce 2,280 conditional states. Volume-only, one-round impact, and multiround-feedback capacity bounds coincide in every state, giving 0% extra reduction in Fig. 24. The maximum initial loss, 1.79158%,

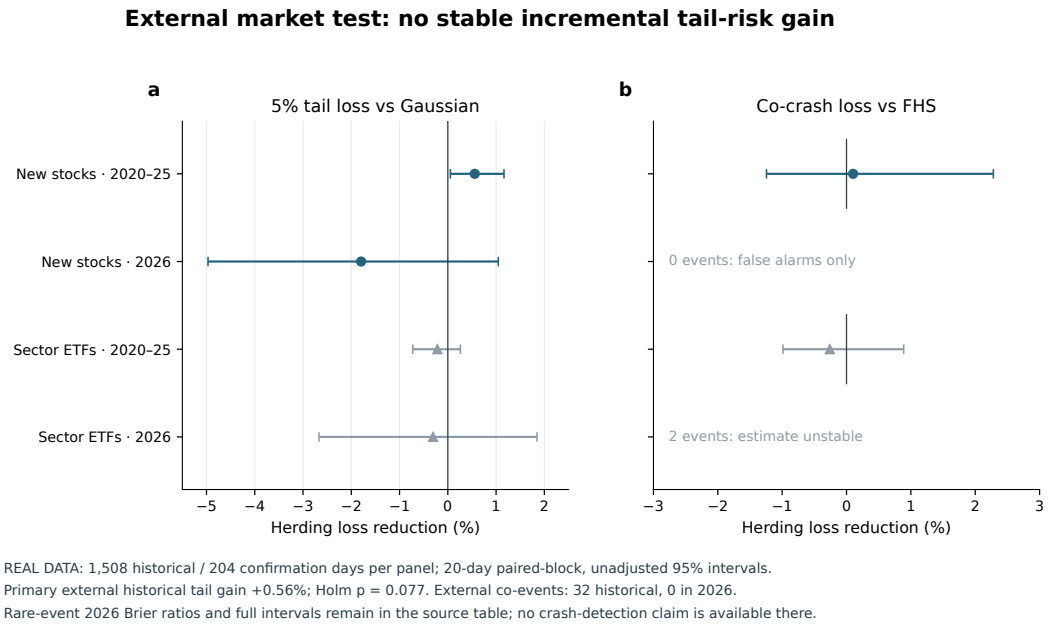


Figure 23. Tail and joint-downside comparisons at matched information and covariance. Statistical references, shrinkage, integration checks and criticality ablations are retained. No co-crash events occur in the 2026 sample, so that period evaluates false alarms. [T03]

is below the fixed 2% trigger. A certificate also verifies nonactivation at smaller scales. The fixed trigger is inactive throughout this design, which makes this branch a certificate of nonactivation under the tested self-impact scenario.

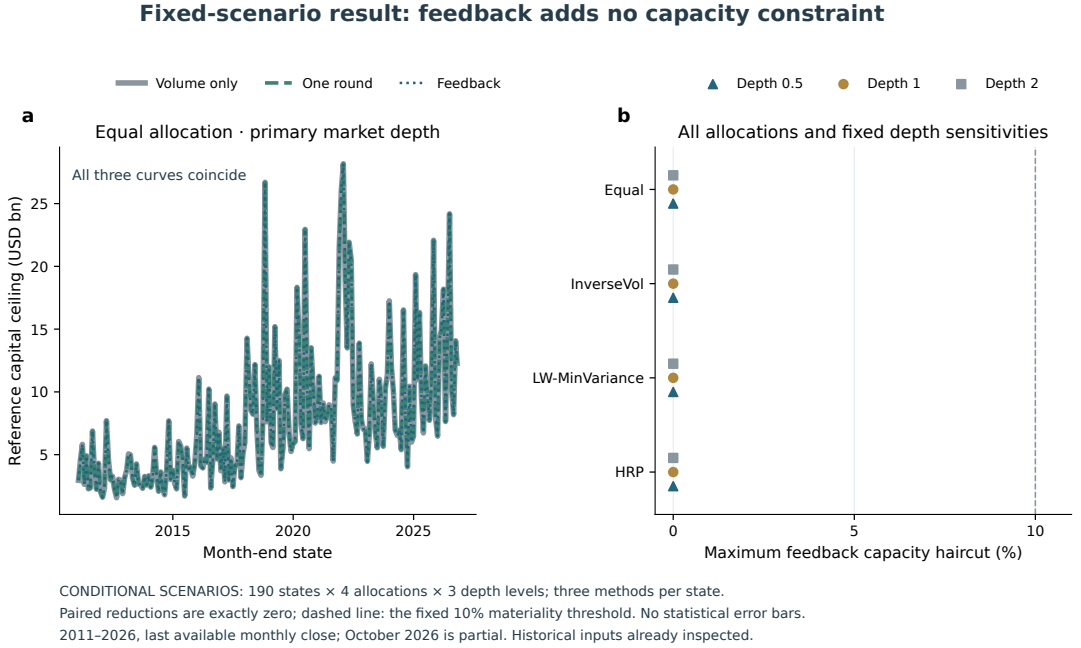


Figure 24. Coincident capacity bounds in the fixed account self-impact scenario. Historical prices and volumes supply conditional inputs; the feedback rule is specified rather than calibrated from execution data. All 2,280 states remain in the source record. [C01]

A conventional volume proxy still exposes a useful trade-off. Among 189 Equal-allocation historical replays with three complete future days, mean-volume sizing exceeds the participation proxy 142 times, or 75.13%. Using the past-20-day 20th-percentile dollar volume reduces this to 62 cases, or 32.80%, while mean deployed capital falls to 72.90%. These historical replays compare capital utilization with subsequent participation-proxy exceedances for a conventional volume rule, separately from the inactive feedback scenario.

The package includes full code, nine capacity CSVs, cash and monotonicity checks, C01–C03 plotting code, and the original certificate.

F Research record and reproduction

F.1 Finding the evidence

The public repository and research website link the shared implementation in `src/econophysics/`, configurations, numerical tables in `results/tables/`, source manifests and three executable notebooks. Bracketed figure identifiers retain the original F/D/R/P/U/V/T/C/J mapping; named continuation figures document the weighted dynamic extension. Table 12 lists the reproduction entry points.

Table 12. Reproduction profiles from the repository root. Each run writes new outputs.

Profile	Operation	Evidence
quick	Exact theory, model and source-table checks	Standard-library mathematics and numerical identities
theory	Exact identities and certificates	Calculations under the stated model assumptions
notebooks	Execute all three research notebooks	Model moments, inversion, market tests and forecast comparisons
site	Check local resources and the browser engine	Website links and agreement with Python fixtures
forecast	Download documented archives into a fresh directory and rebuild the fixed experiment	All 183 historical test dates and 16 saved CSV tables
Extended laboratory	Rebuild retained experiment pages	Risk, strategy overlap, allocation and coordination diagnostics

F.2 Dynamic continuation evidence

The public package in `studies/collective_risk_continuation_v3/` contains numerical source tables, original hashes, protocols, independent arithmetic checks and the Explore data export. It distinguishes source-law simulations, the conditional-ensemble design and market-transport diagnostics. Primary ensemble splits are 2,048 bank, 32 tuning, 64 calibration and 128 evaluation trajectories. The earlier particle-model calibration is a separate design; its scores are not pooled with this evaluation.

F.3 Environment and source records

Forecast reconstruction uses Python 3.14.4 and the versions in `requirements-study.txt`; plotting and notebook dependencies are listed separately. A fresh reconstruction reproduces all 16 saved forecast tables byte for byte. Notebooks retain executed outputs, and the quick and website checks pass on Python 3.11 and 3.14.

Binance Vision-derived results retain CC BY-NC-SA 4.0 attribution. Full supplier price histories are acquired through the source manifests. Code, data and third-party asset terms are documented in `LICENSE.md` and `DATA_LICENSES.md`. Historical experimental digests remain in the source record.

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